Lecture 20 (11/2)

Residual plots and feature maps

Today:

1) Residual plots to identify interactions and other model deficiencies
2) Feature maps and polynomial regression

Residual plots

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The residuals are what the fitted model leaves unexplained:

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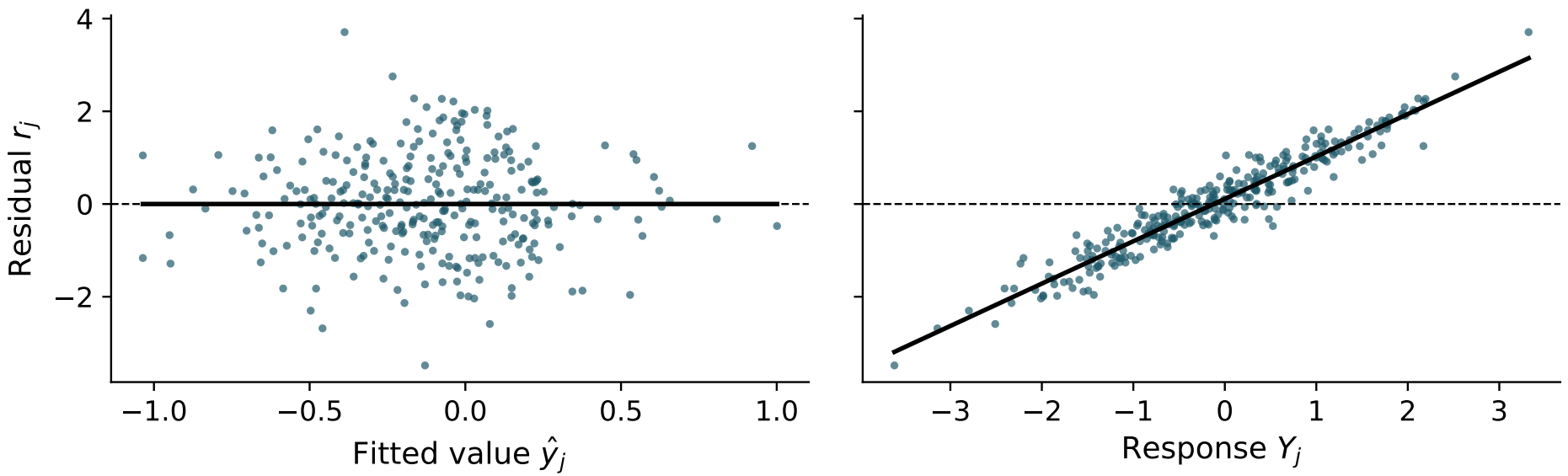
(r_j is computed from the data; ϵ_j in the model cannot be observed.)

If the model is right, r_j is approximately Normal$(0, \sigma_\epsilon^2)$ *whatever the values of the predictors*. A residual plot is meant to visualize the r_j.

Residuals vs. fitted values and vs. Y

Consider

$$Y = 0.5X + \epsilon, \quad \sigma_\epsilon = 1$$



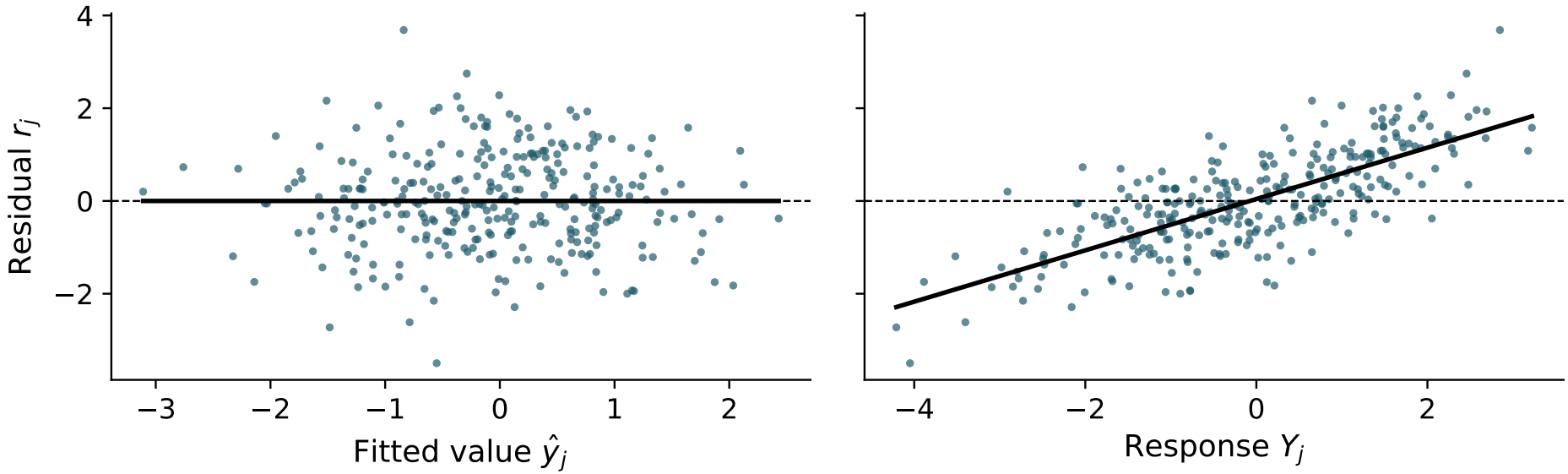
Question

Why is there a trend on the right plot?

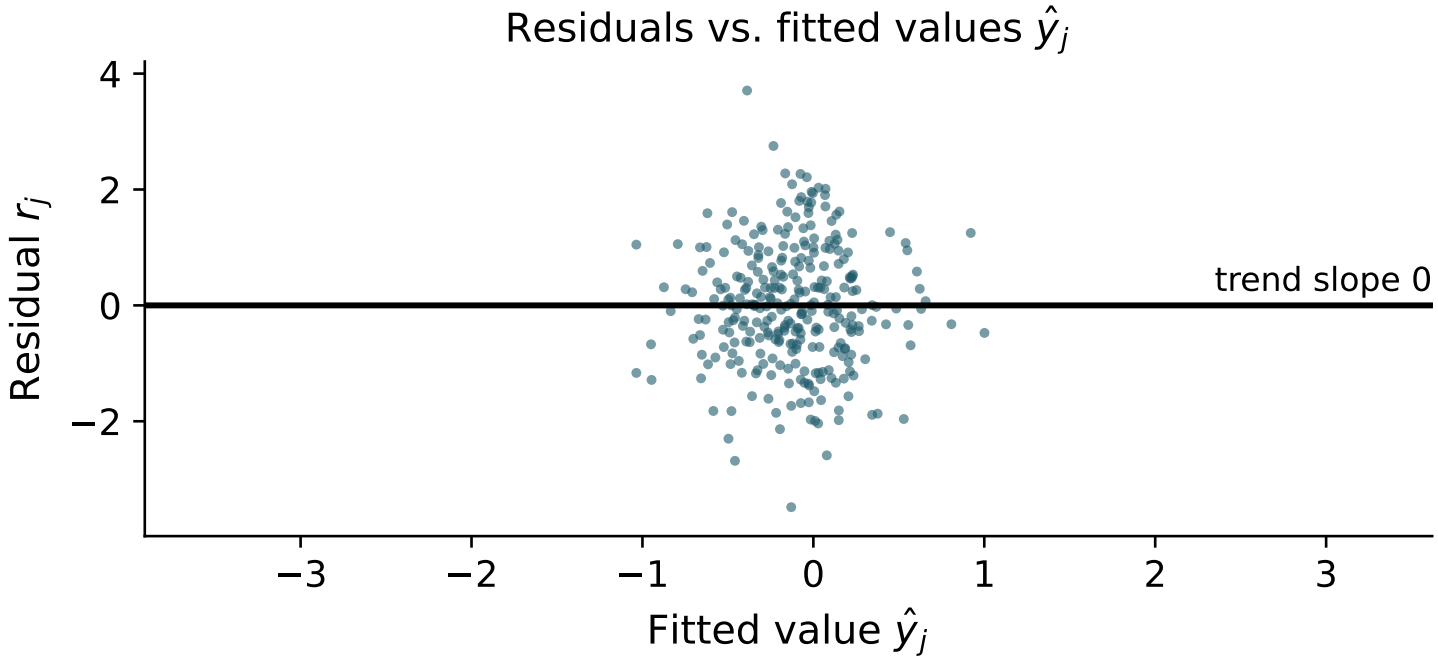
Residuals vs. fitted values and vs. Y: two predictors

Consider

$$Y = 0.2X_1 + 20X_2 + \epsilon, \quad \sigma_\epsilon = 1$$

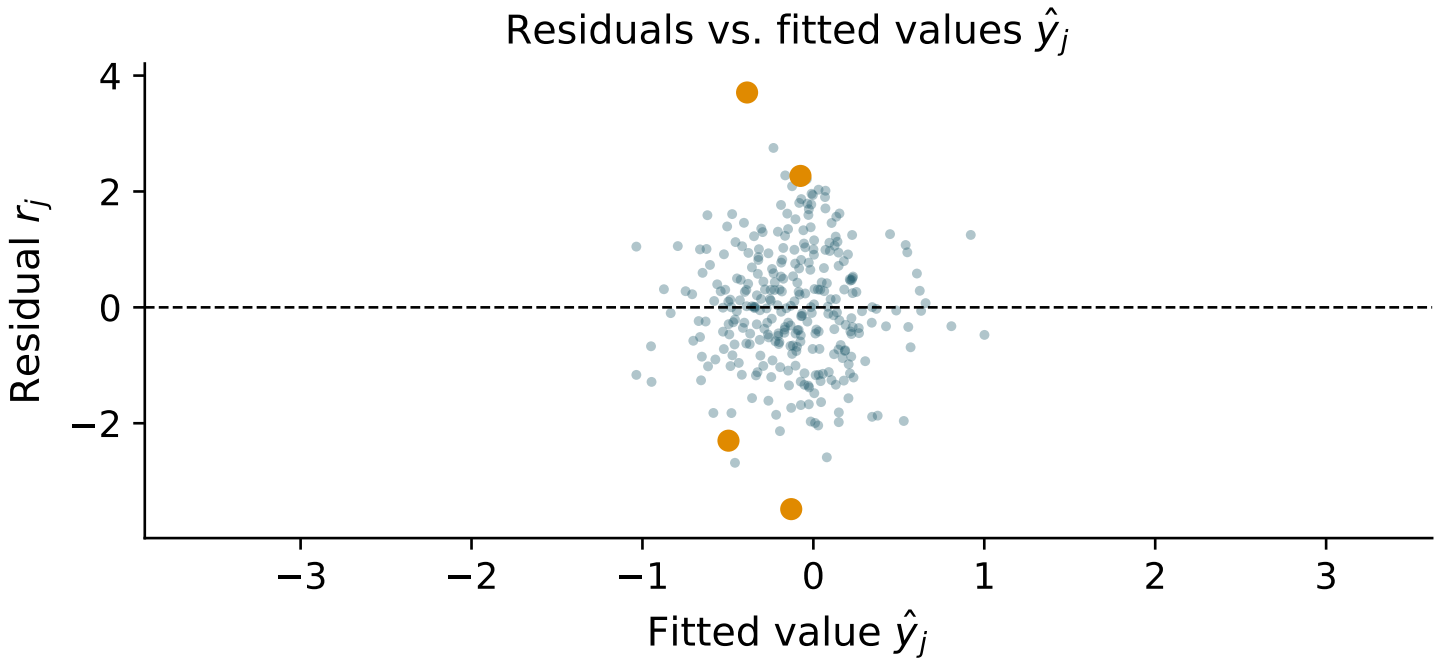


Why plot $\hat{y}_i$ on x axis



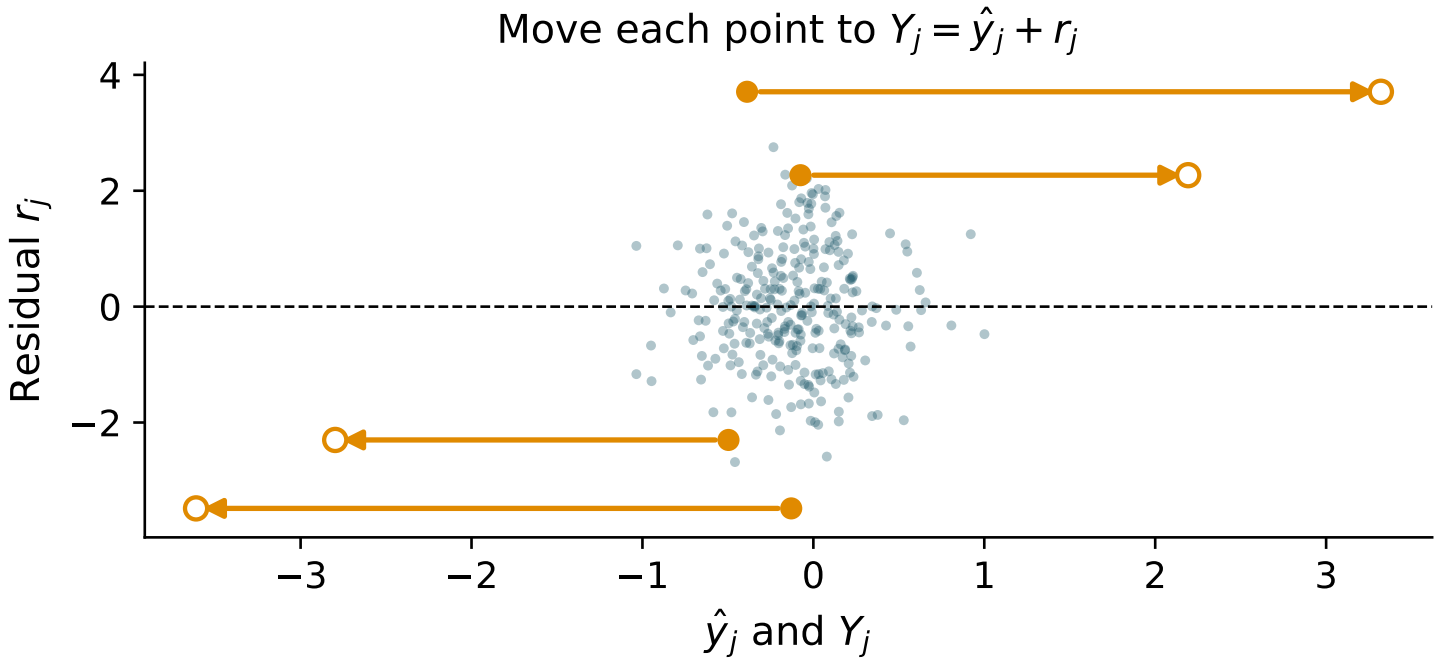
1. If the model is right, r_j has the same distribution whatever the predictors are, so vs. $\hat{y}_j$ we expect a symmetric spread in the vertical direction.

Why plot $\hat{y}_i$ on x axis



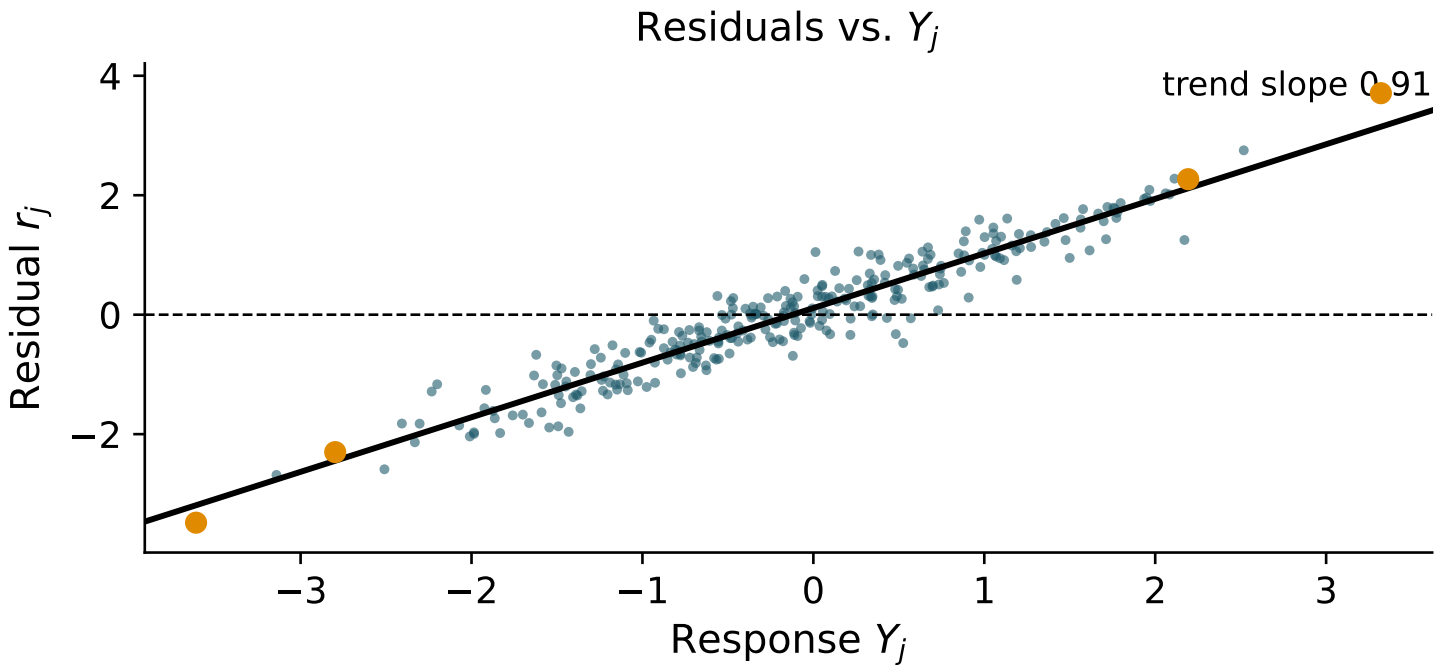
2. Pick out a few points with large residuals.

Why plot $\hat{y}_j$ on x axis



3. Since $Y_j = \hat{y}_j + r_j$, putting Y_j on the horizontal axis shifts each point sideways by the corresponding residual.

Why plot $\hat{y}_i$ on x axis



4. Every point shifts this way, so the cloud tilts even though the model is correct.

Example

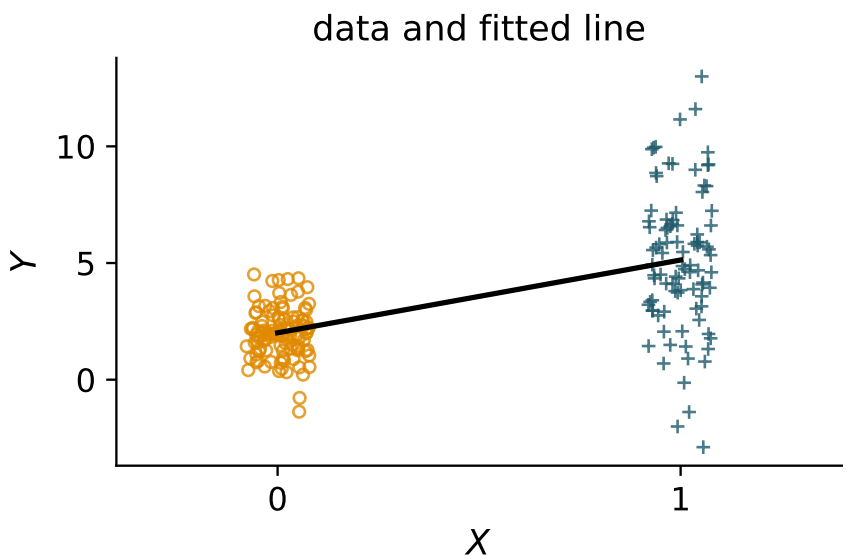
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$$Y = 2 + 3X + \epsilon(1 + 2X)$$

Below is the data with the fitted line.

Question

What will the residual plot look like when the data is fit to a linear regression model?



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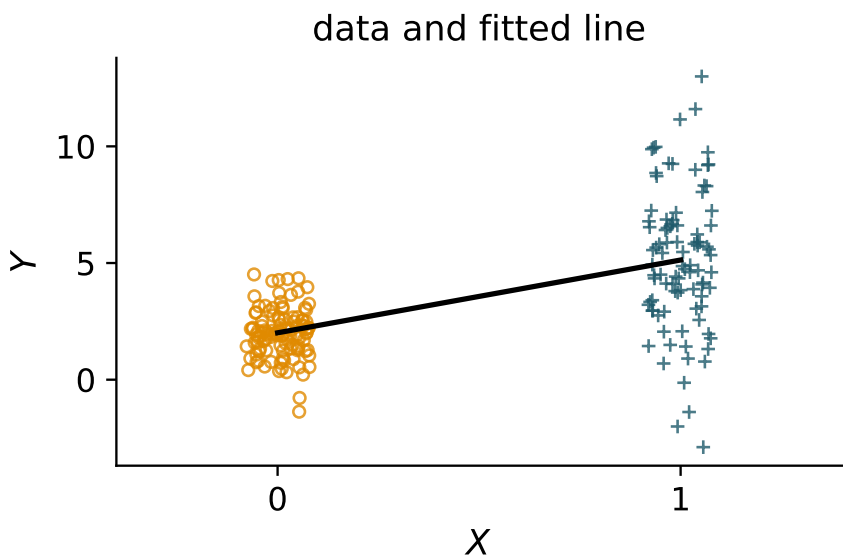
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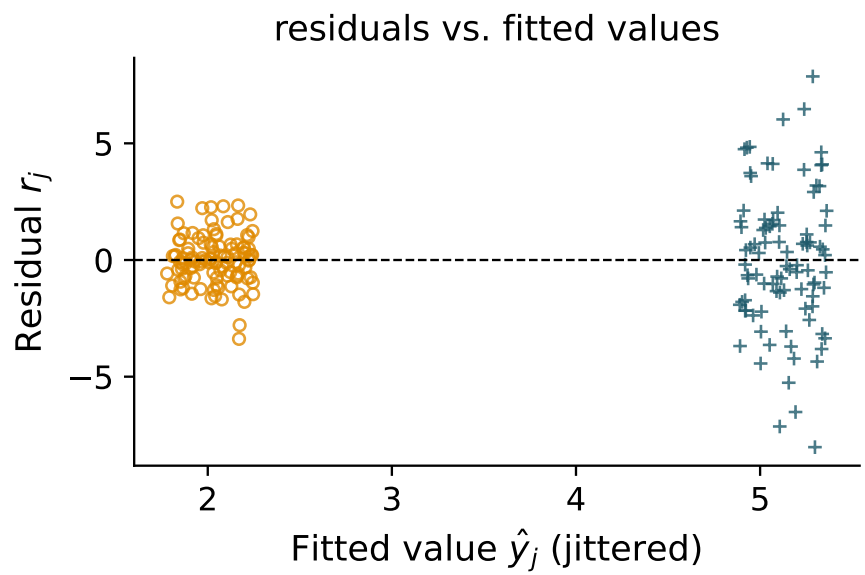
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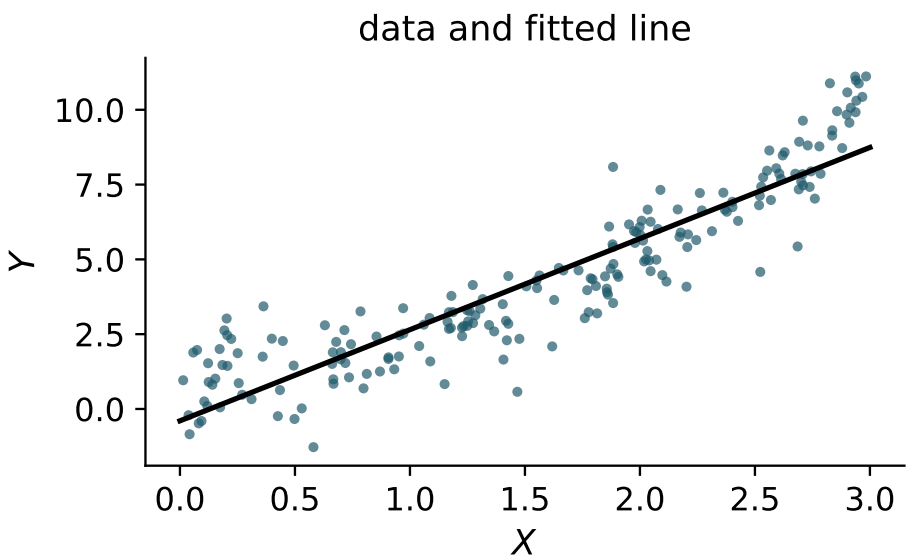
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$$Y = 1 + X^2 + \epsilon$$

where X is between 0 and 3. Below is the data with the fitted line.

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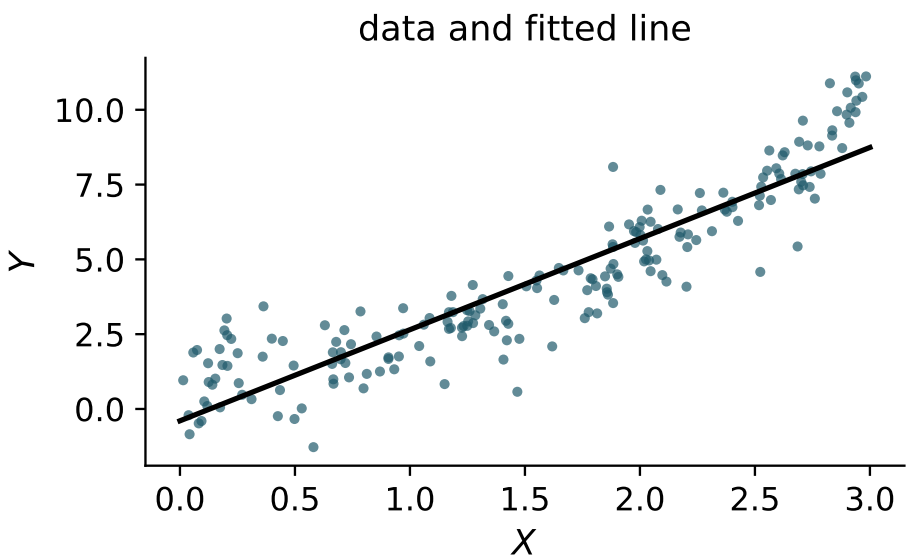
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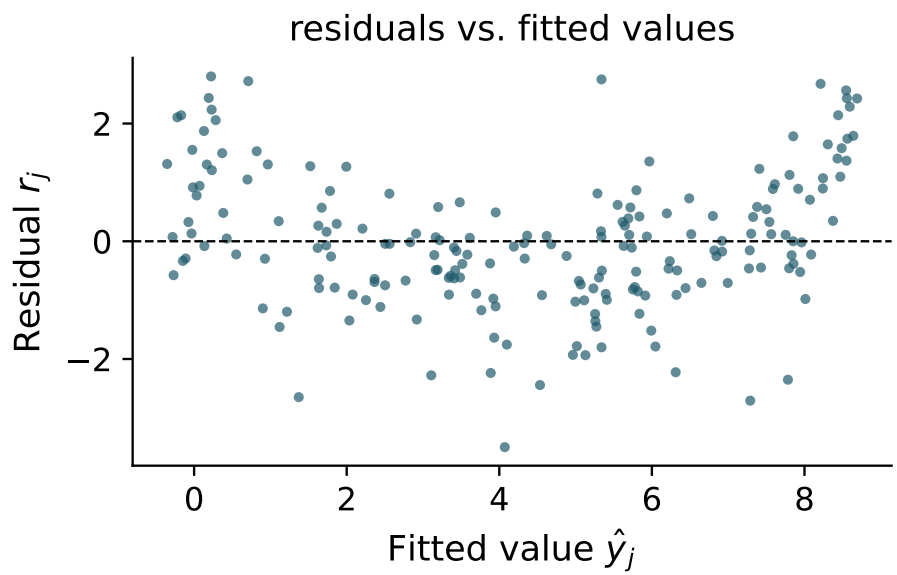
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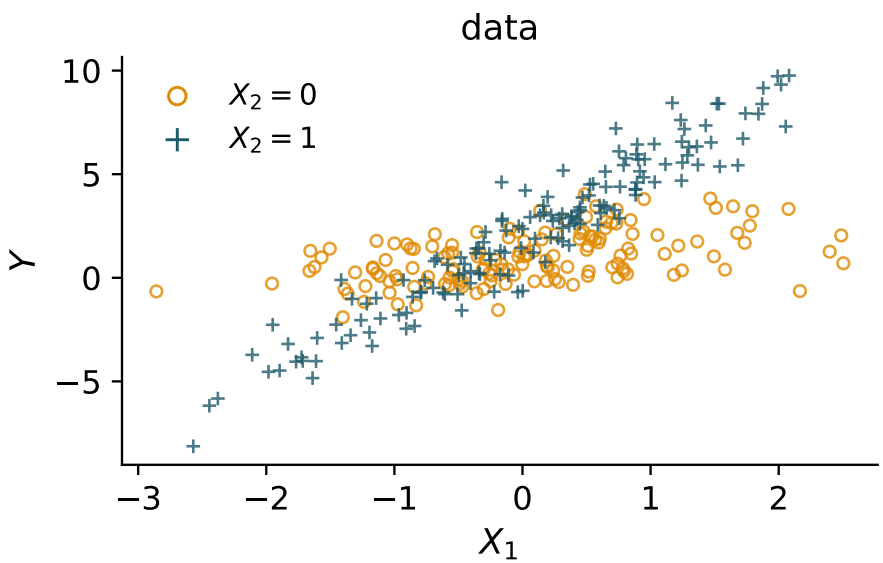
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$$Y = 1 + 0.5X_1 + X_2 + 3X_1X_2 + \epsilon$$

where X_2 is a binary predictor. Below is the data.

Question

Sketch the residual plot when this model is fitted without the interaction term.



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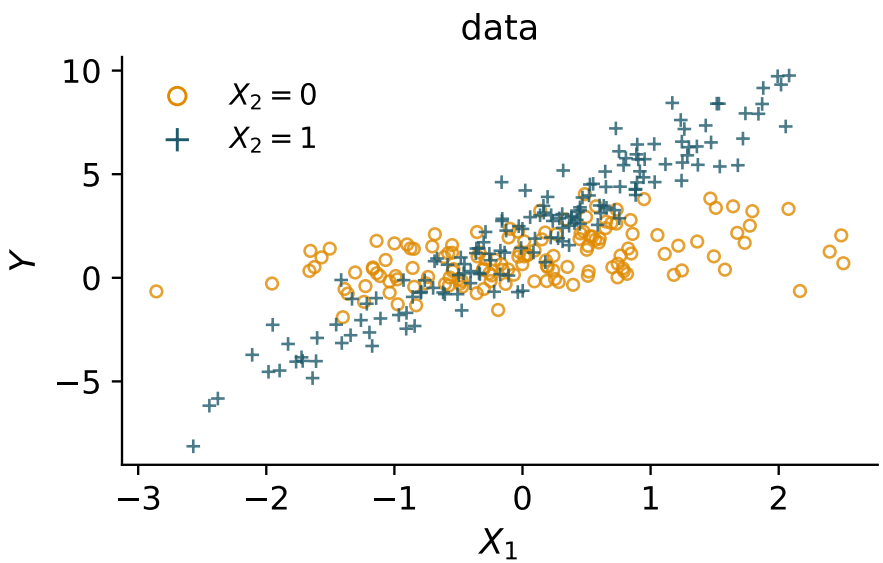
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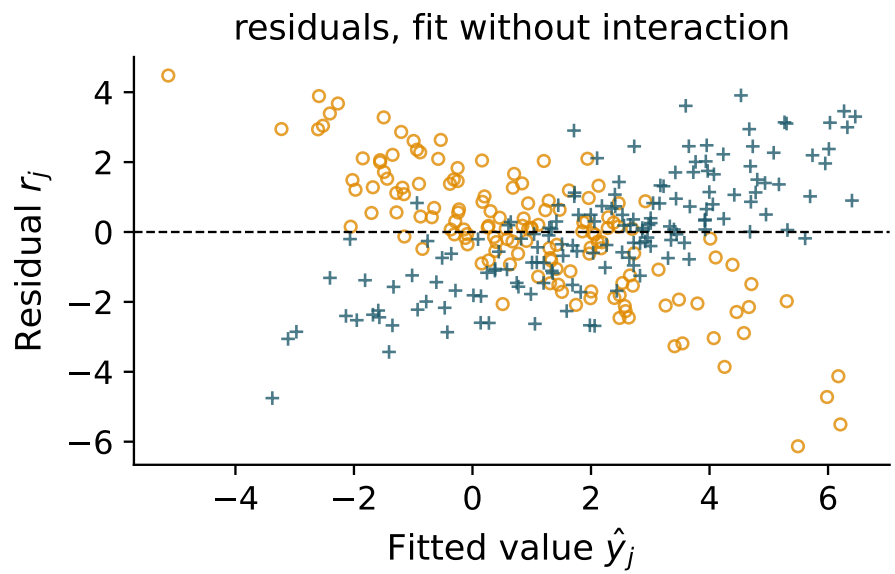
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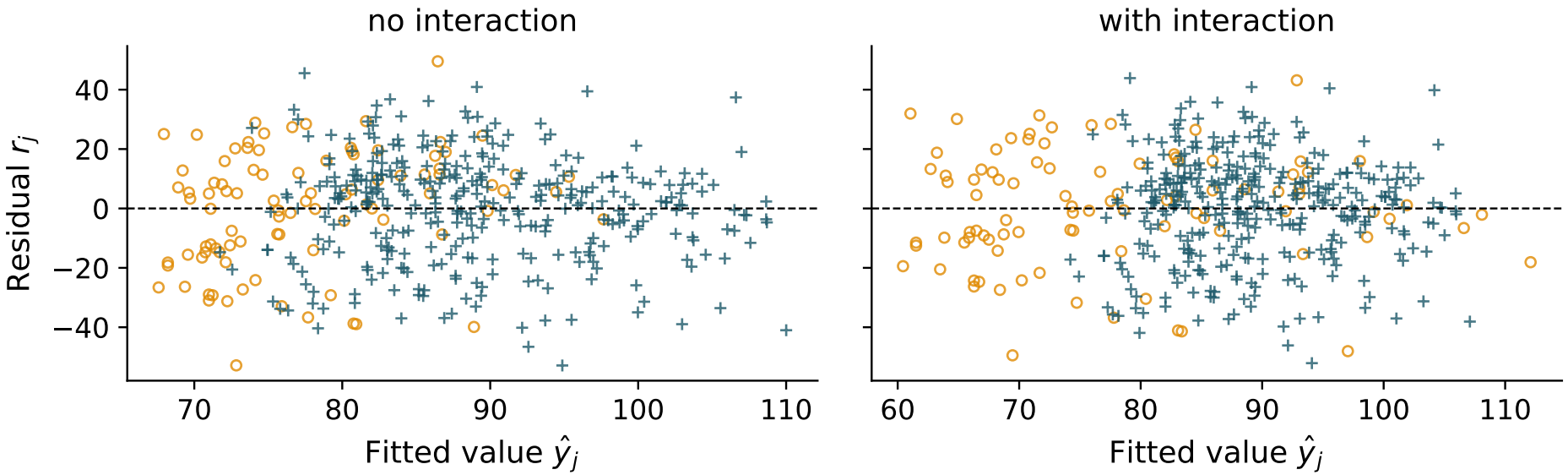
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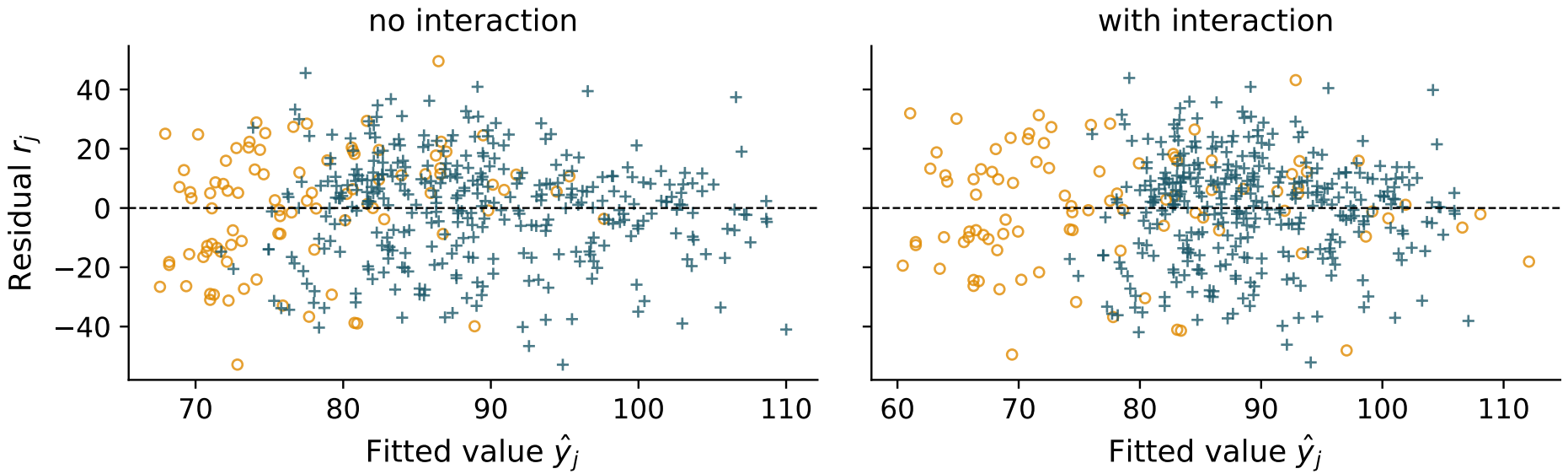




Test scores: residuals vs. fitted values



Test scores: residuals vs. fitted values



Question

The missing interaction is not evident from the residual plot, but it is statistically significant ($p = 0.003$). What does this suggest about how R^2 changes when adding the residual?

Example: advertising budgets and sales

Sales of a product in 200 markets (from *An Introduction to Statistical Learning*): Y is sales in thousands of units, and X_{TV}, X_{radio} are the TV and radio advertising budgets in \$1000s. Without an interaction:

Dep. Variable:		sales	R-squared:			0.897
	coef	std err	t	P>|t|	[0.025	0.975]
const	2.9211	0.294	9.919	0.000	2.340	3.502
TV	0.0458	0.001	32.909	0.000	0.043	0.048
radio	0.1880	0.008	23.382	0.000	0.172	0.204

With an interaction, and both budgets centered at their means (147.0 and 23.3):

Dep. Variable:		sales	R-squared:			0.968
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const	13.9470	0.067	208.737	0.000	13.815	14.079
TV_c	0.0444	0.001	56.673	0.000	0.043	0.046
radio_c	0.1886	0.005	41.806	0.000	0.180	0.198
TV_radio_c	0.0011	5.24e-05	20.727	0.000	0.001	0.001

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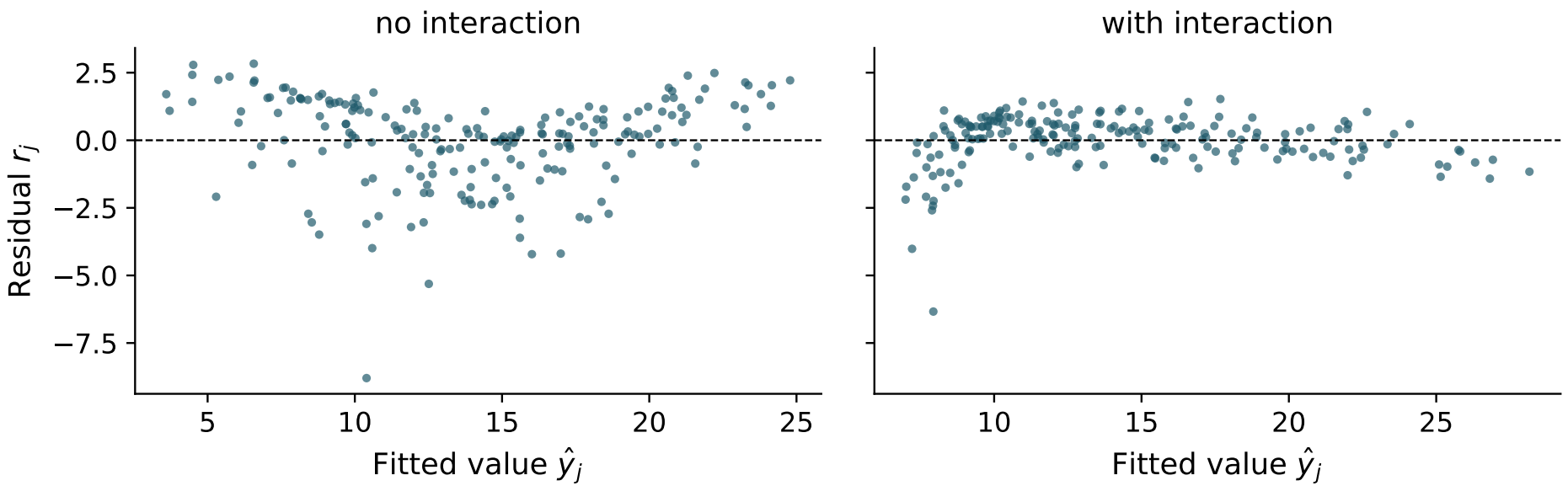
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Question

How does the effect of an extra \$1000 of radio advertising on sales depend on the TV budget?

Advertising: residuals without vs. with interaction



Example: the SARS-CoV-2 spike protein

Moulana et al. (2022) measured how strongly the spike protein binds the human ACE2 receptor for every combination of the 15 mutations in the Omicron BA.1 variant. Y is the log binding affinity of the spike protein to an enzyme ACE2. $X_i = 1$ if mutation i is present.

We use four sites, Q498R, N501Y, Y505H and K417N, and the model with all second-order interactions:

$$Y = \beta_0 + \sum_{i=1}^{4} \beta_i X_i + \sum_{i<k} J_{ik} X_i X_k + \epsilon.$$

This has $1 + 4 + 6 = 11$ coefficients. The other 11 mutations vary in the background and end up in ϵ.

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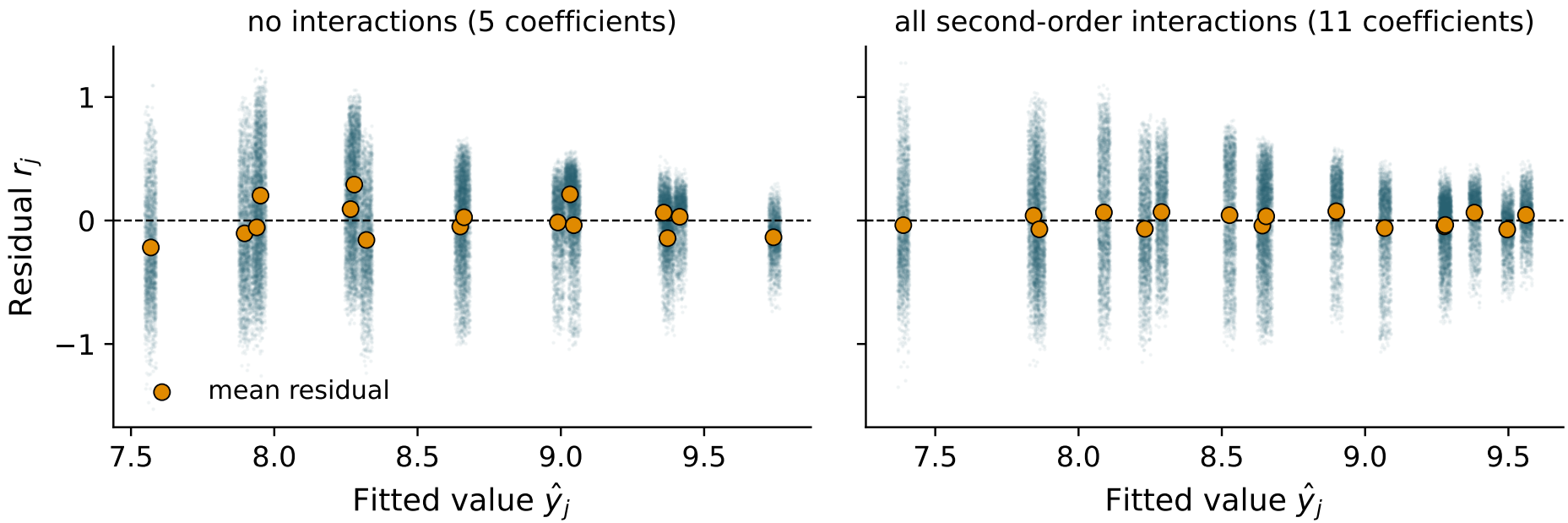
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$R^2 = 0.75$ without interactions (5 coefficients) and 0.78 with them.

Spike protein: residuals vs. fitted values

Each combination of the four sites has one fitted value, so the residuals fall in 16 vertical bands. The orange dots are the mean residual in each band.



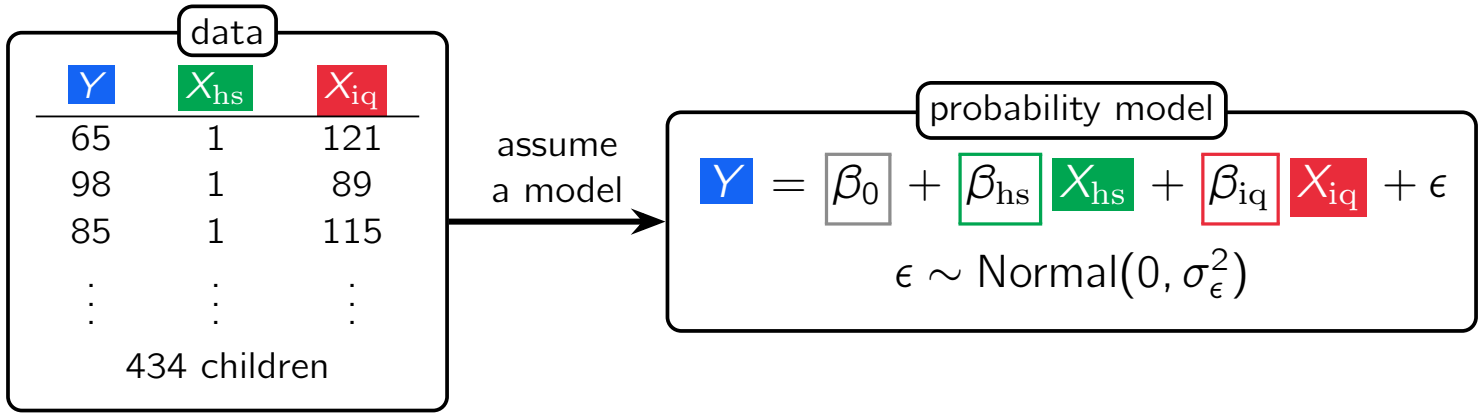
Statstics: Recap

data

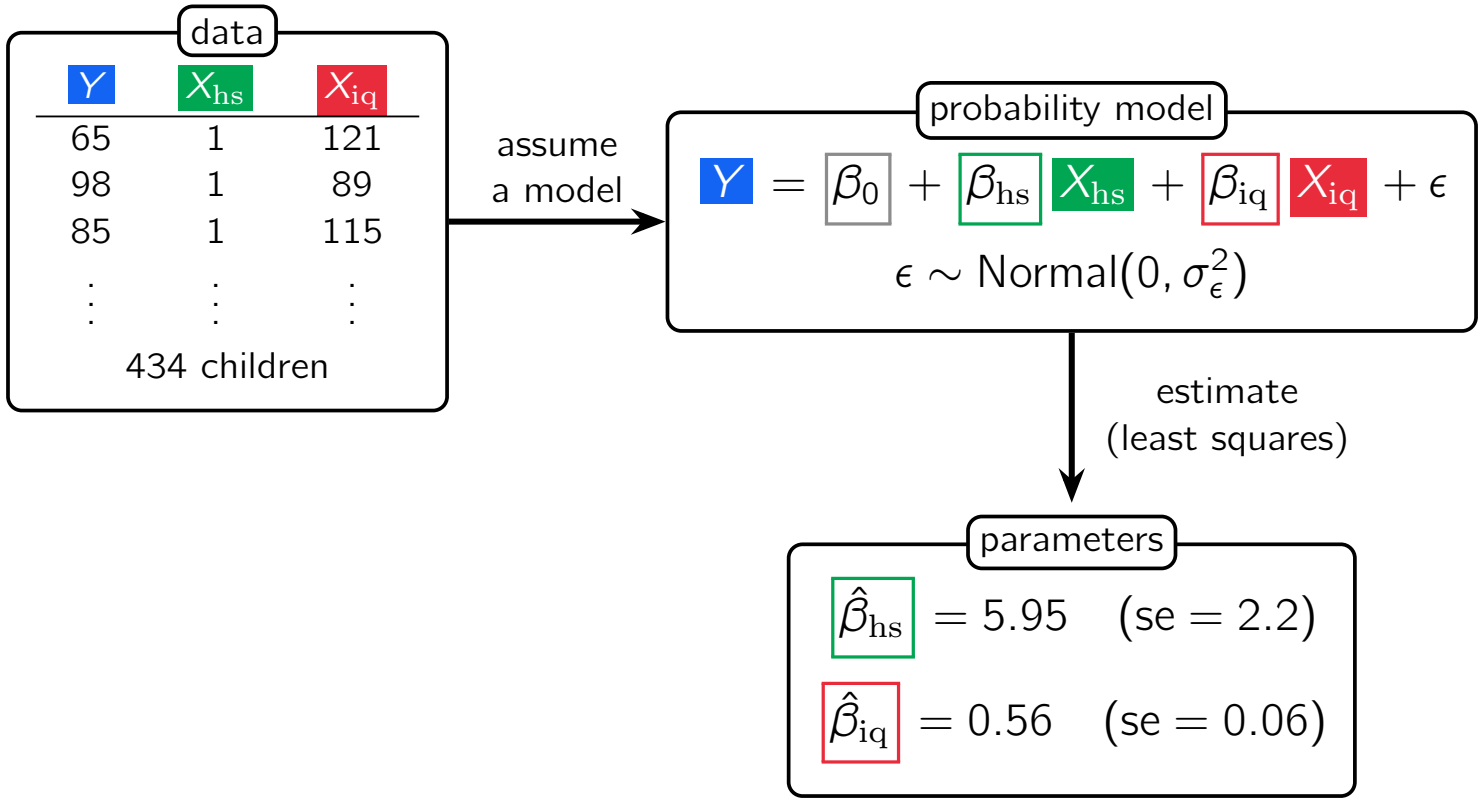
Y	X_{hs}	X_{iq}
65	1	121
98	1	89
85	1	115
⋮	⋮	⋮

434 children

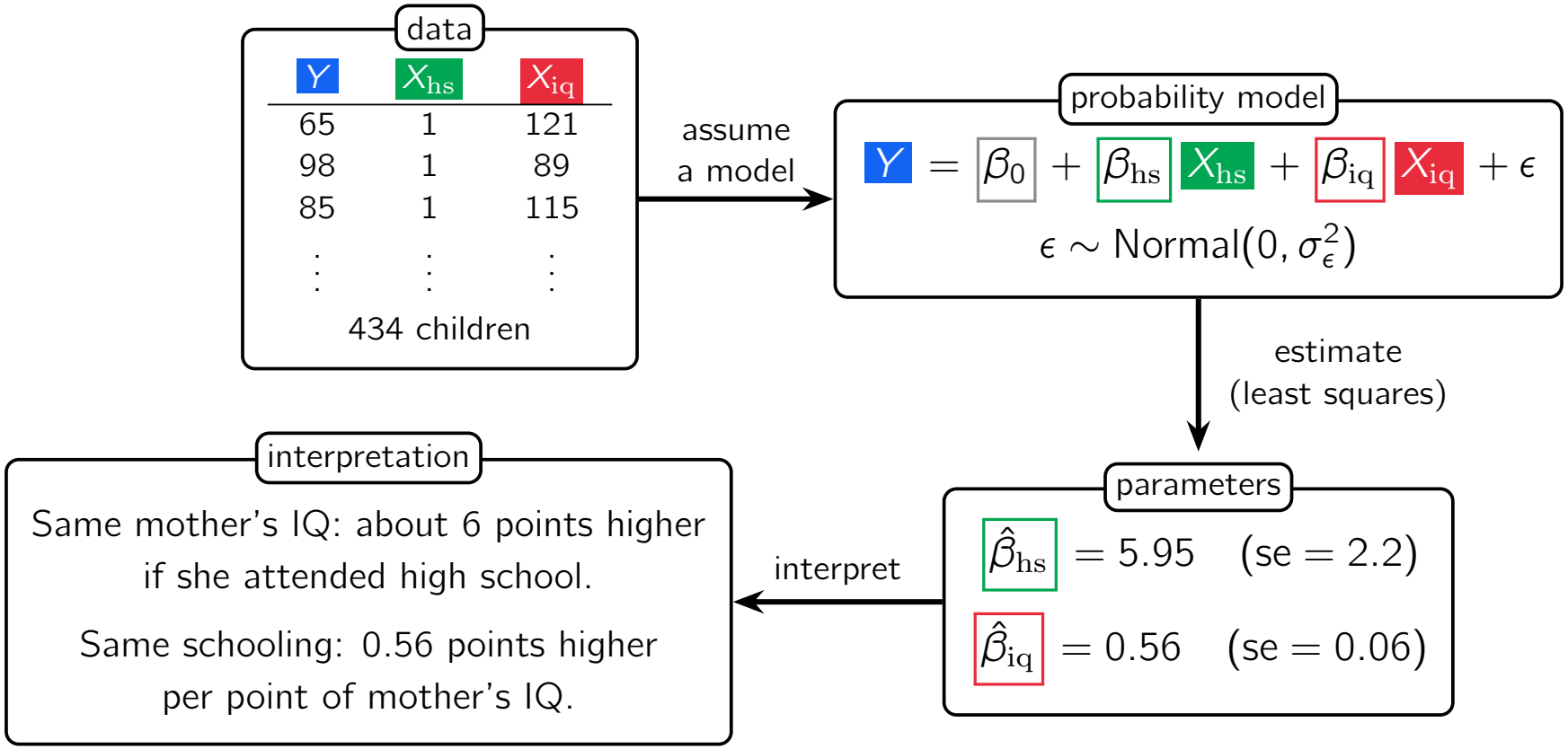
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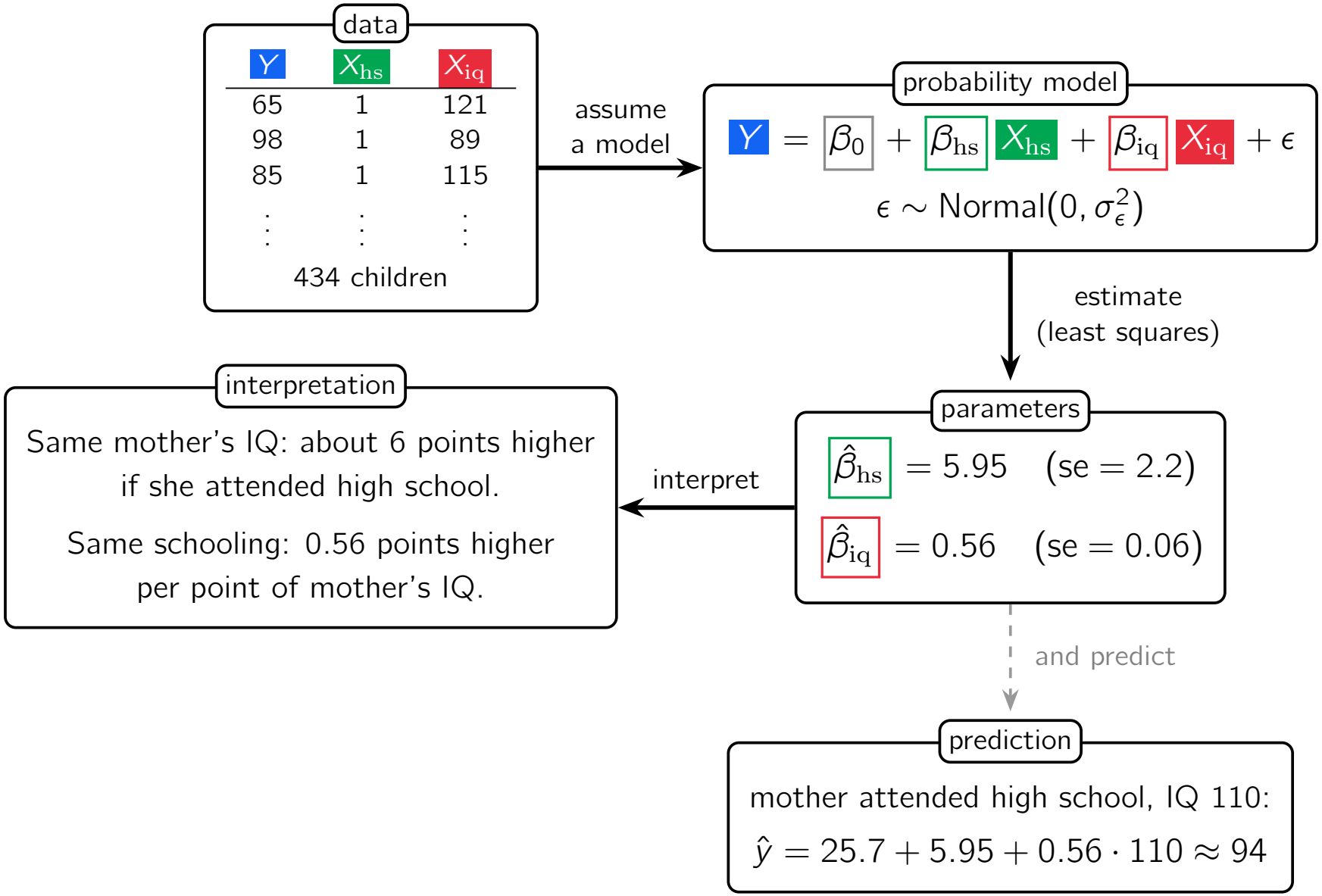
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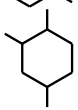
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What does it mean for a model to be complex? Does having more parameters make it more complex? Can a model be too complex?

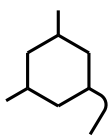
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Example: drug discovery

data: N molecules

molecule	binding energy
	y_1
	y_2
	y_3
⋮	⋮
	y_N

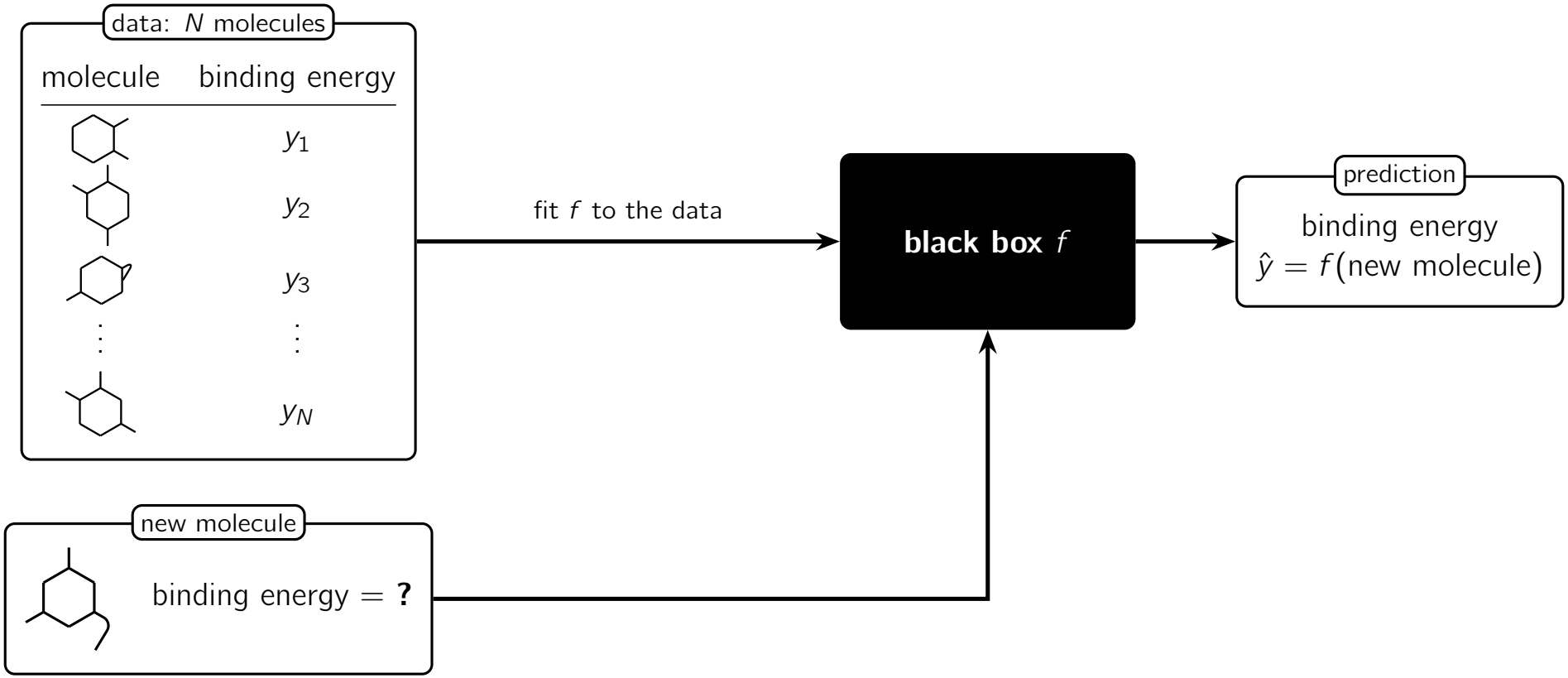
new molecule



binding energy = **?**

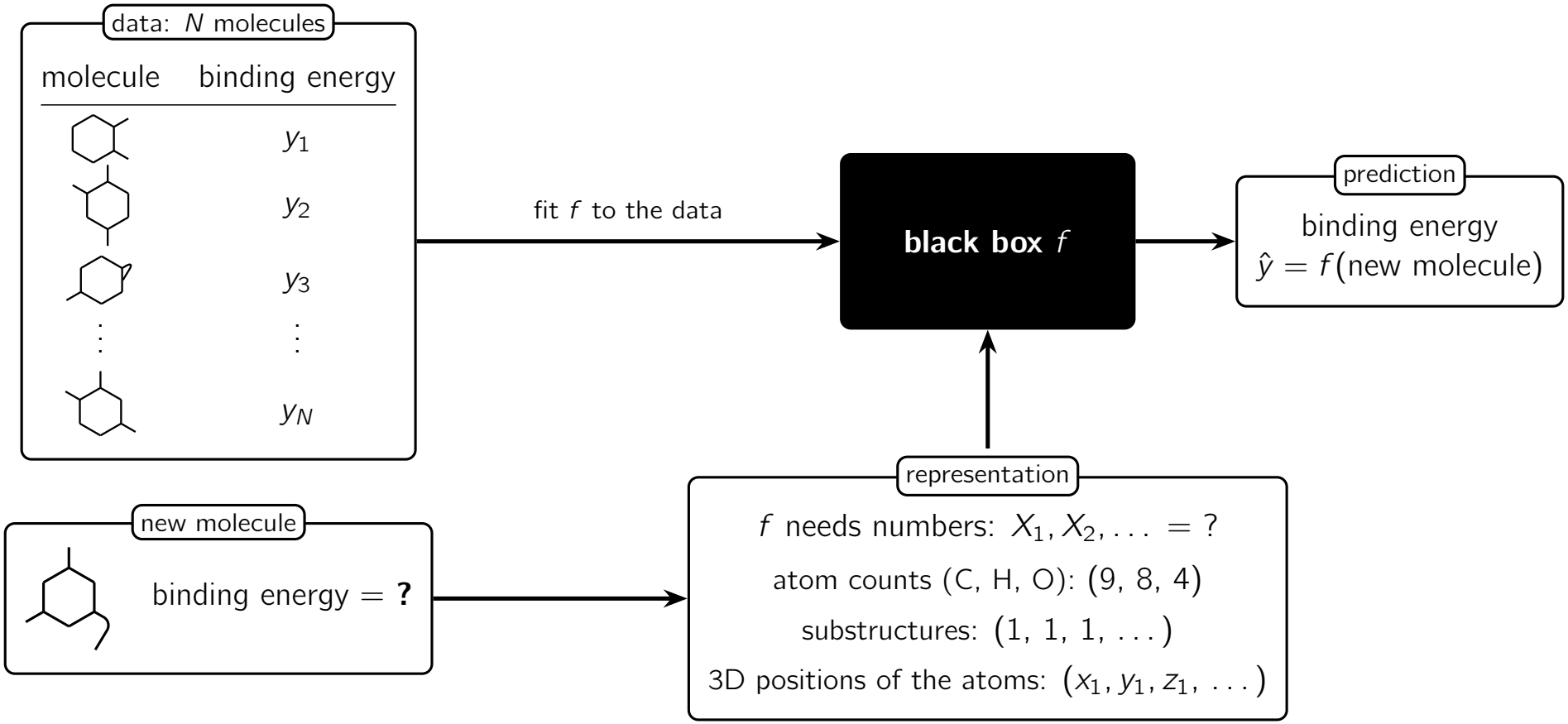
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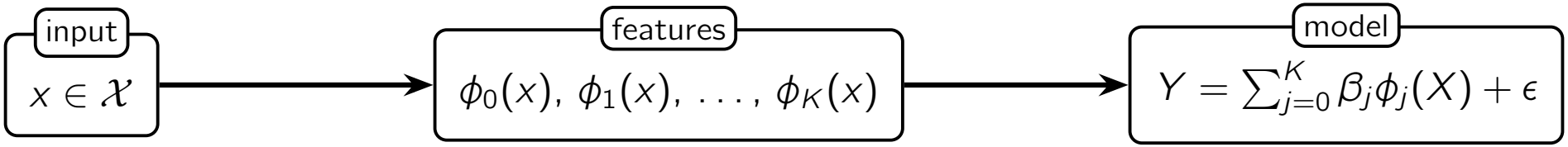
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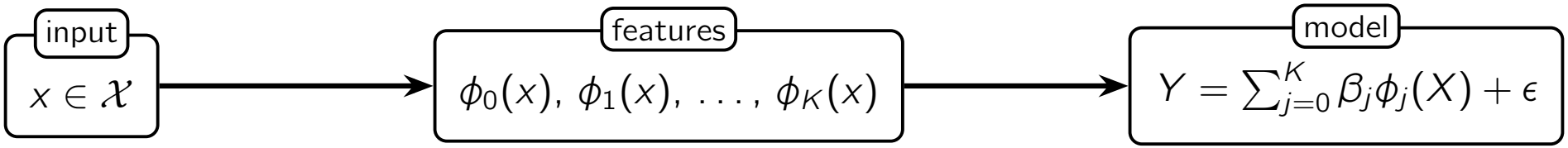
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There is an unknown function f from a space of inputs $\mathcal{X}$ to $\mathbb{R}$, and we observe $Y = f(X) + \epsilon$. We want to learn what f looks like.



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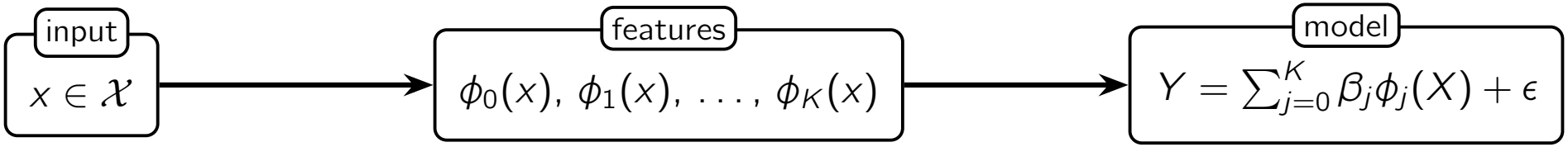
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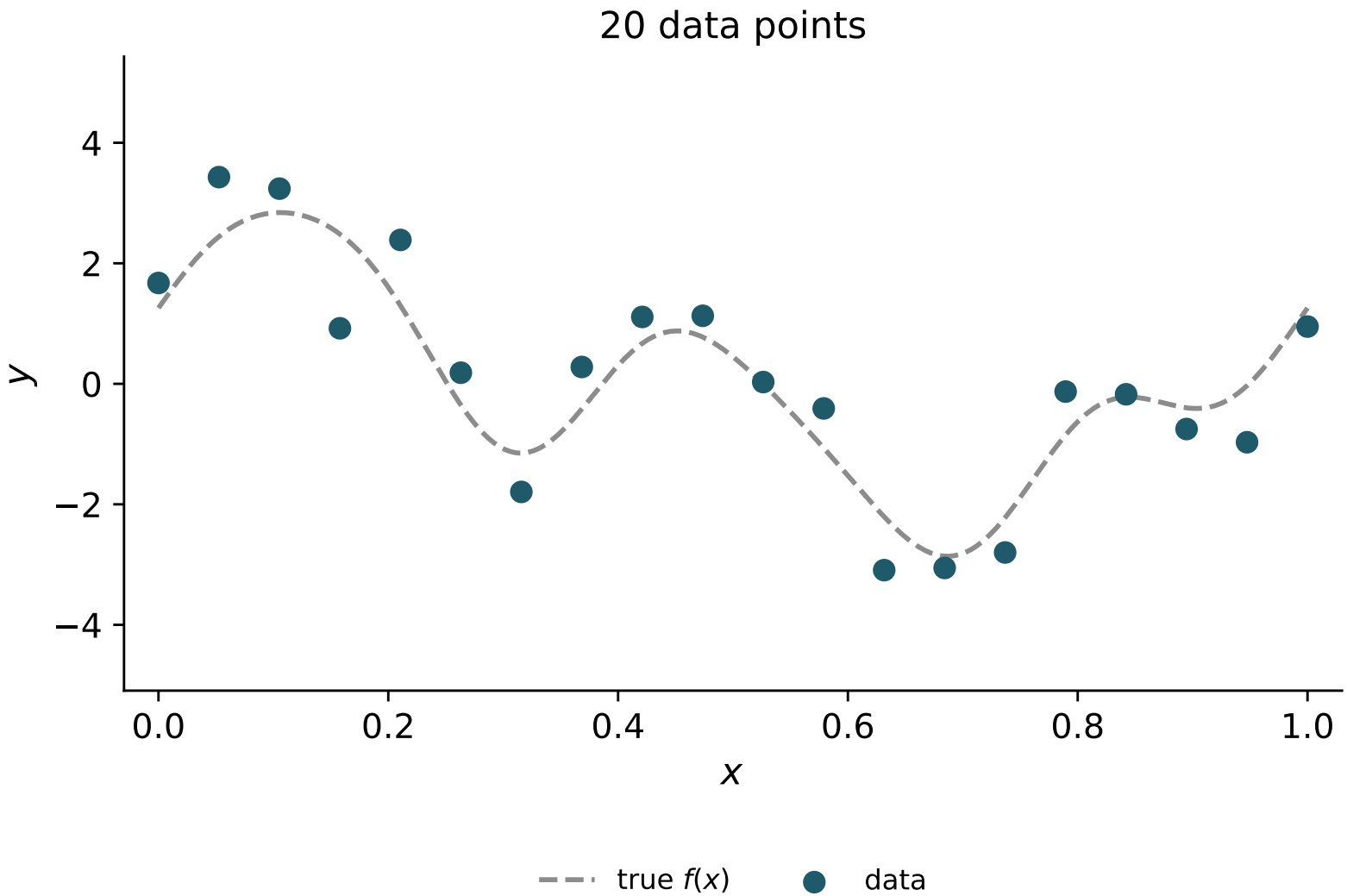


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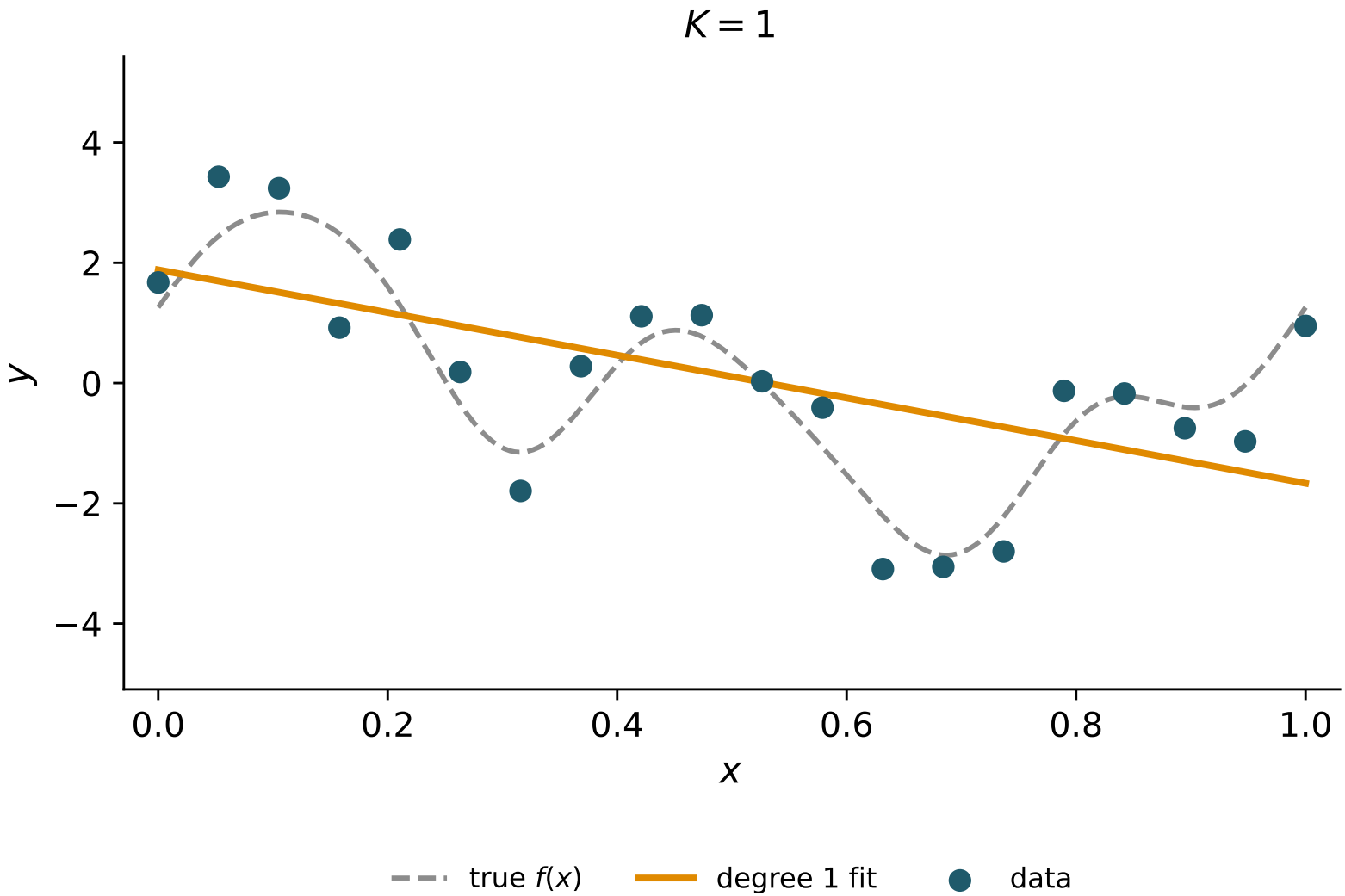
Polynomial regression: $\mathcal{X} = \mathbb{R}$ and $\phi_j(x) = x^j$, so

$$Y = \beta_0 + \beta_1 X + \beta_2 X^2 + \cdots + \beta_K X^K + \epsilon.$$

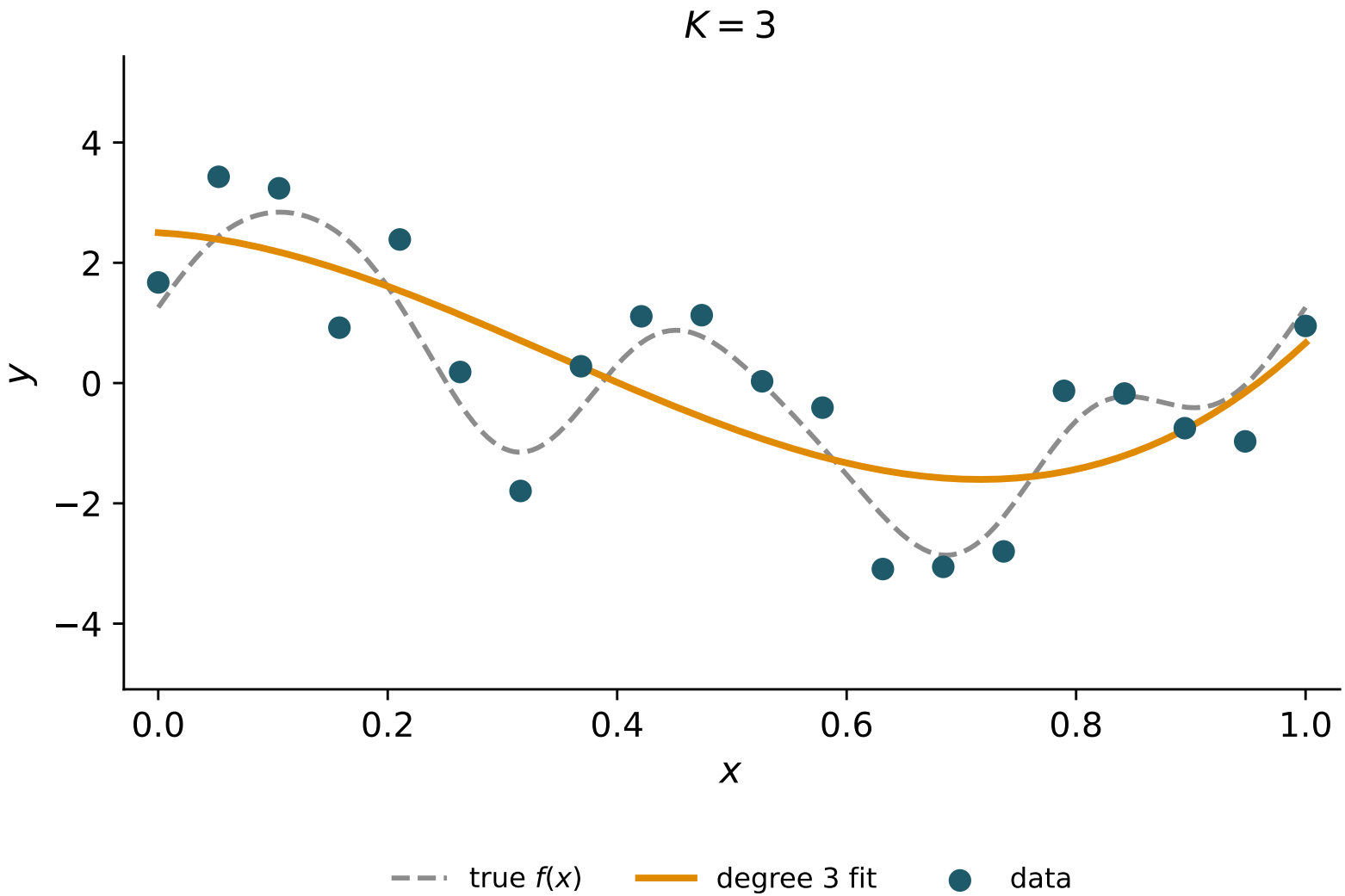
Example: polynomials fitted to a random function



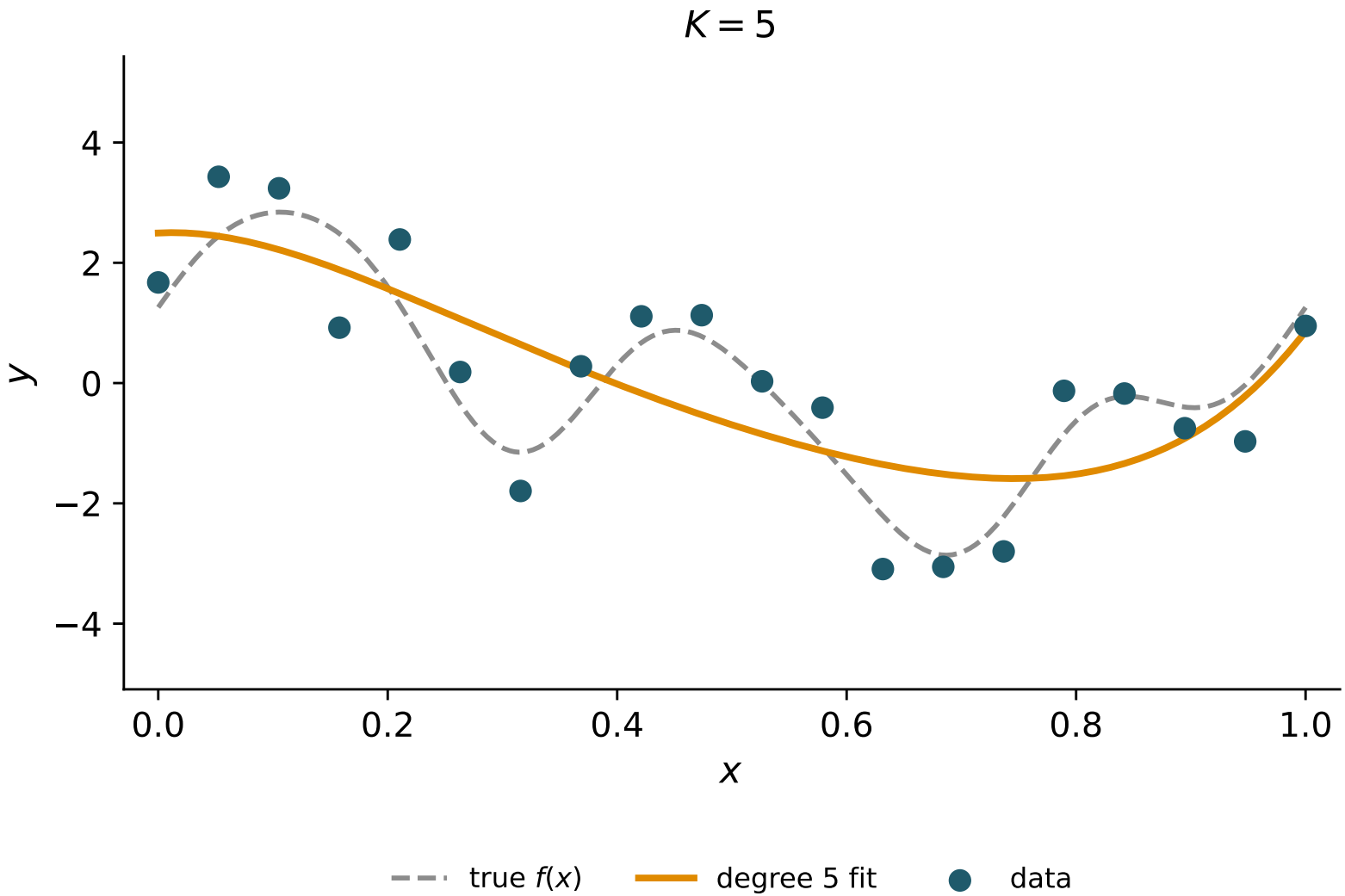
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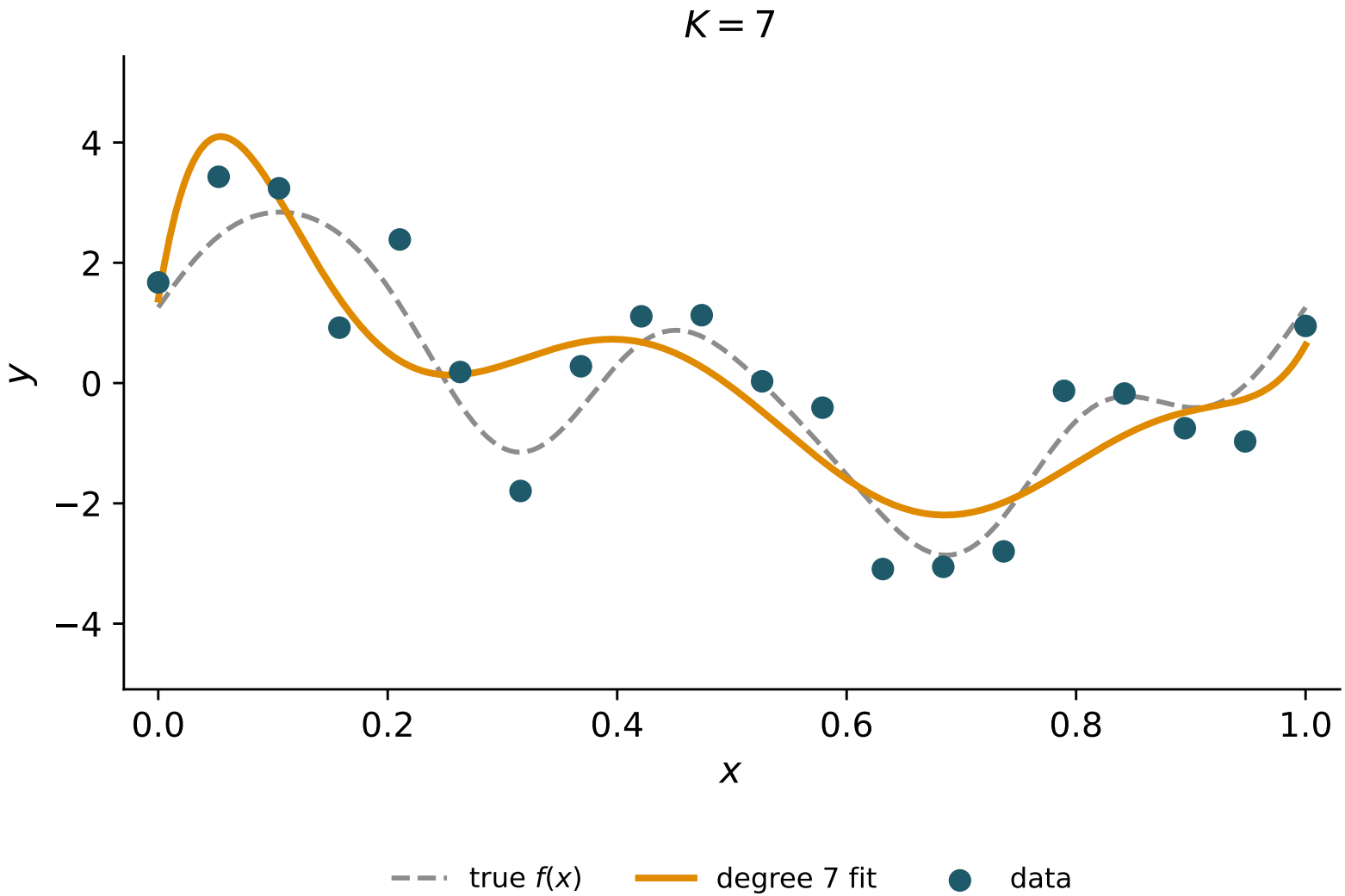
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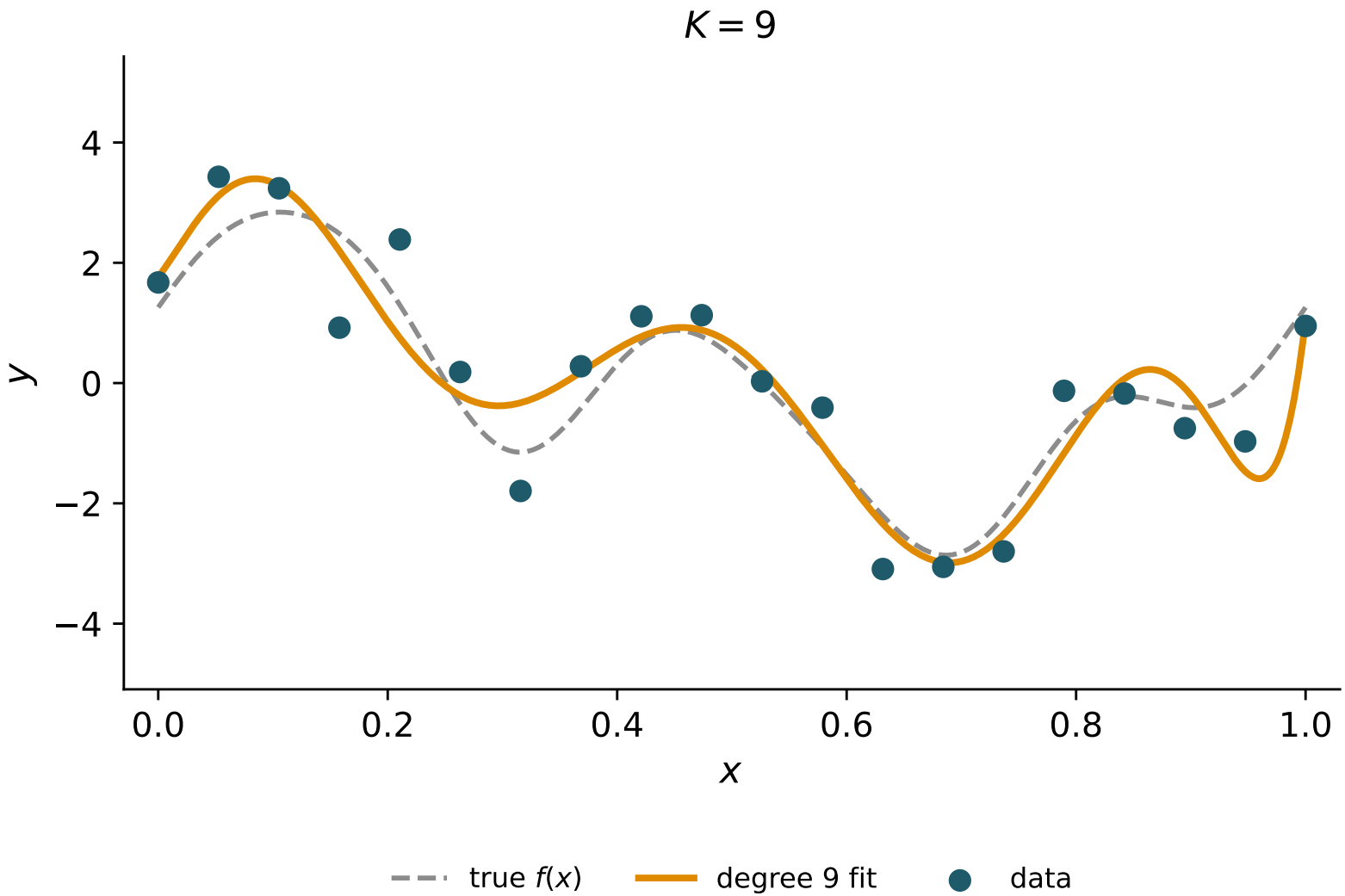
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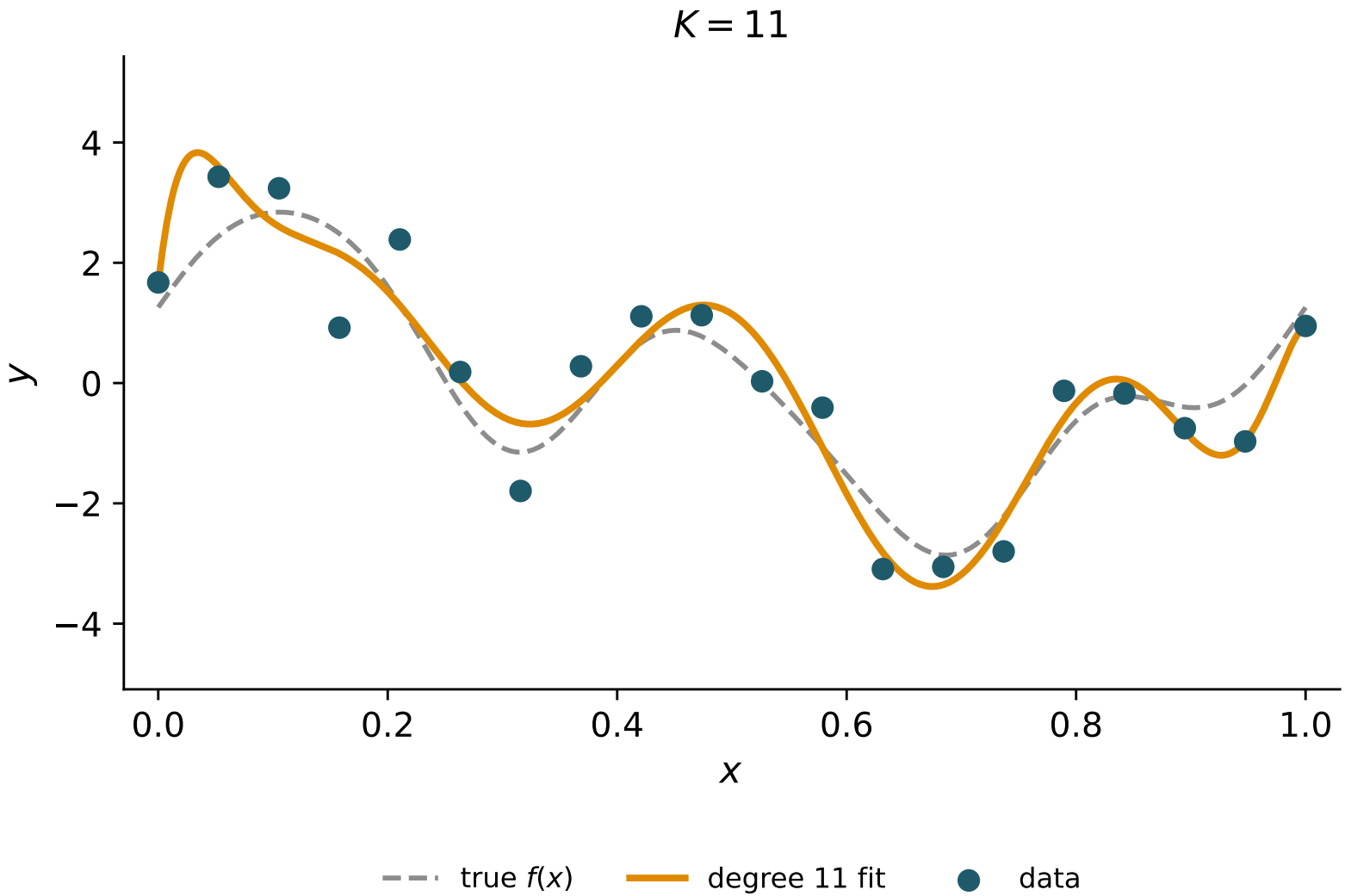
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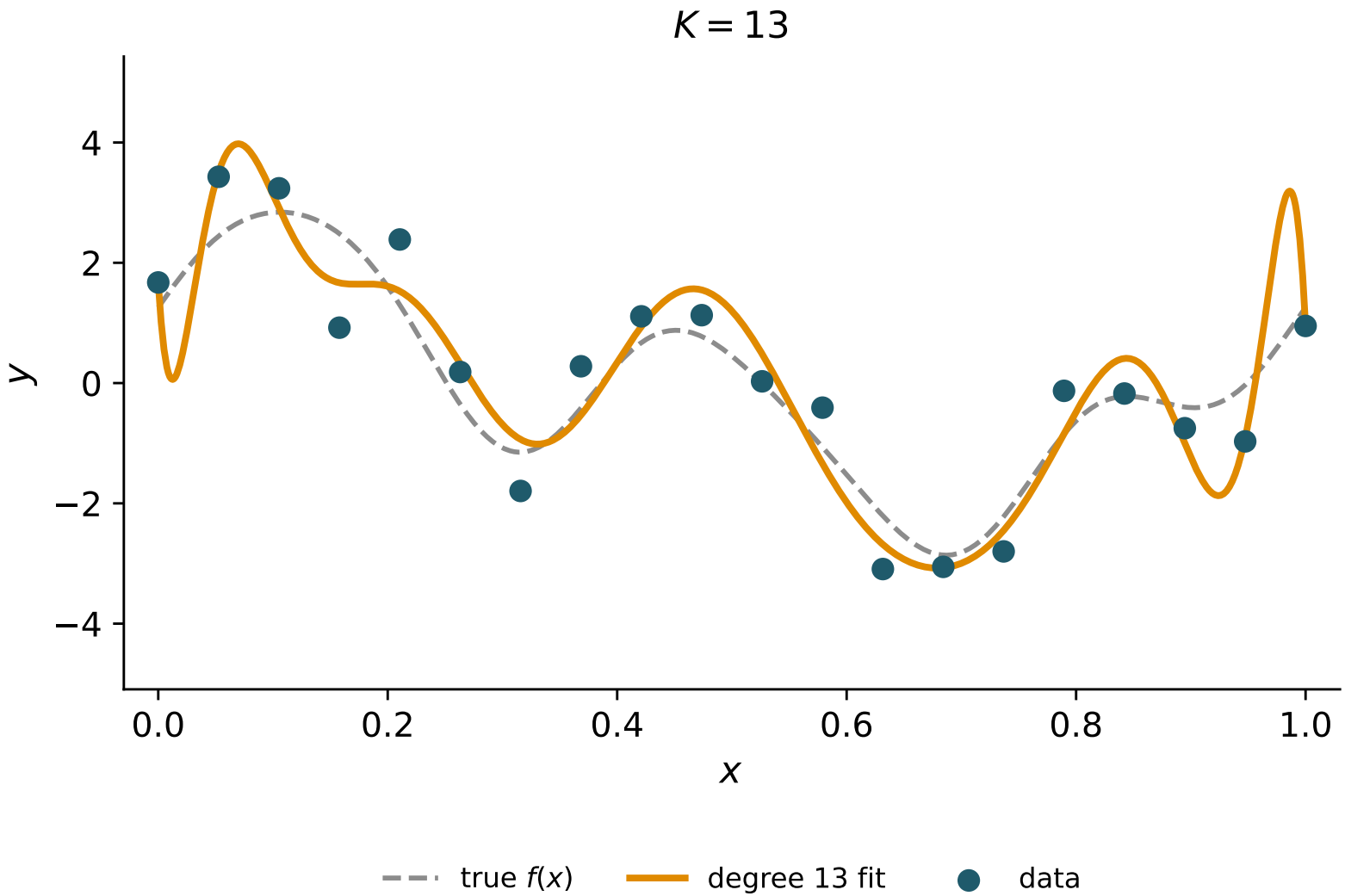
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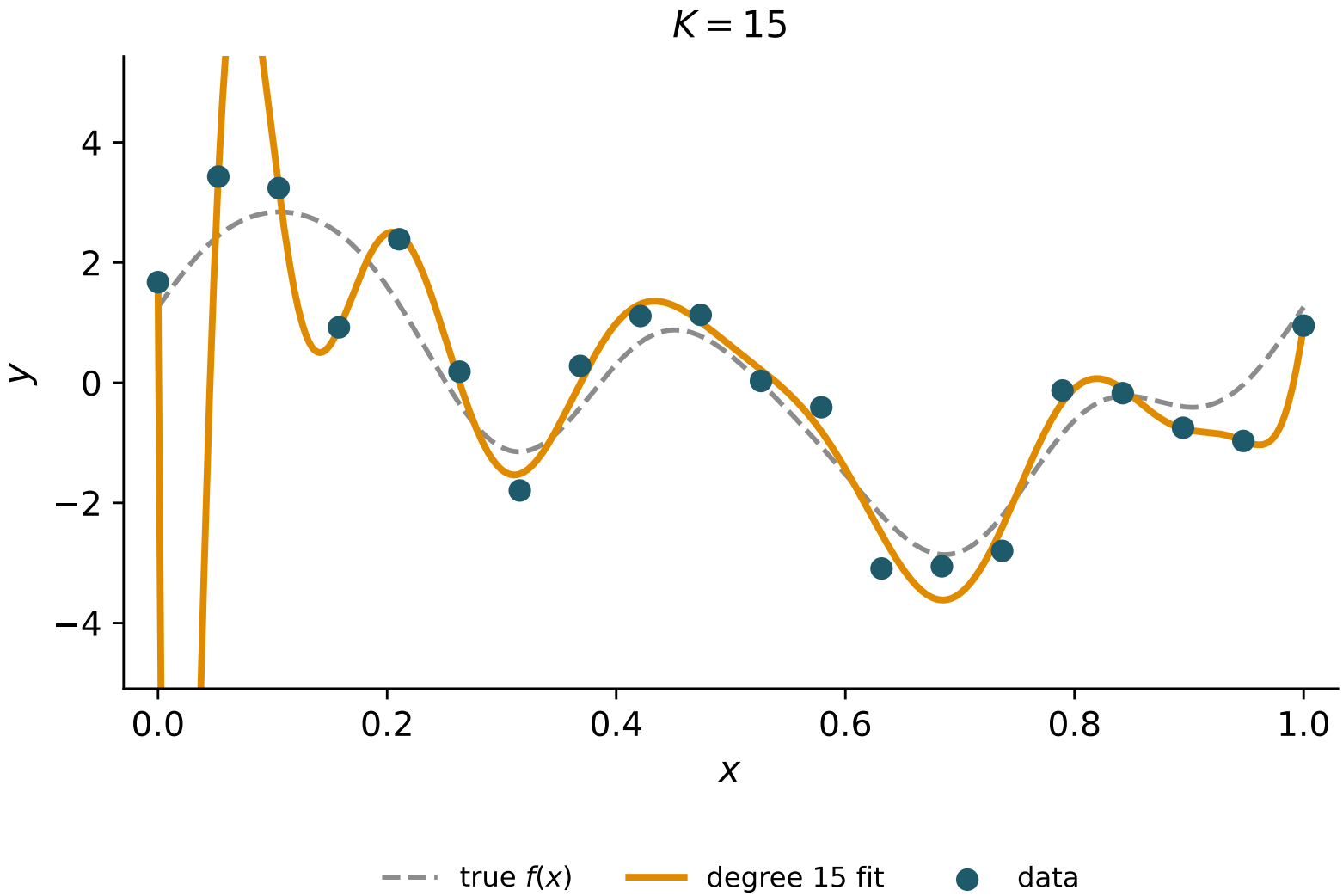
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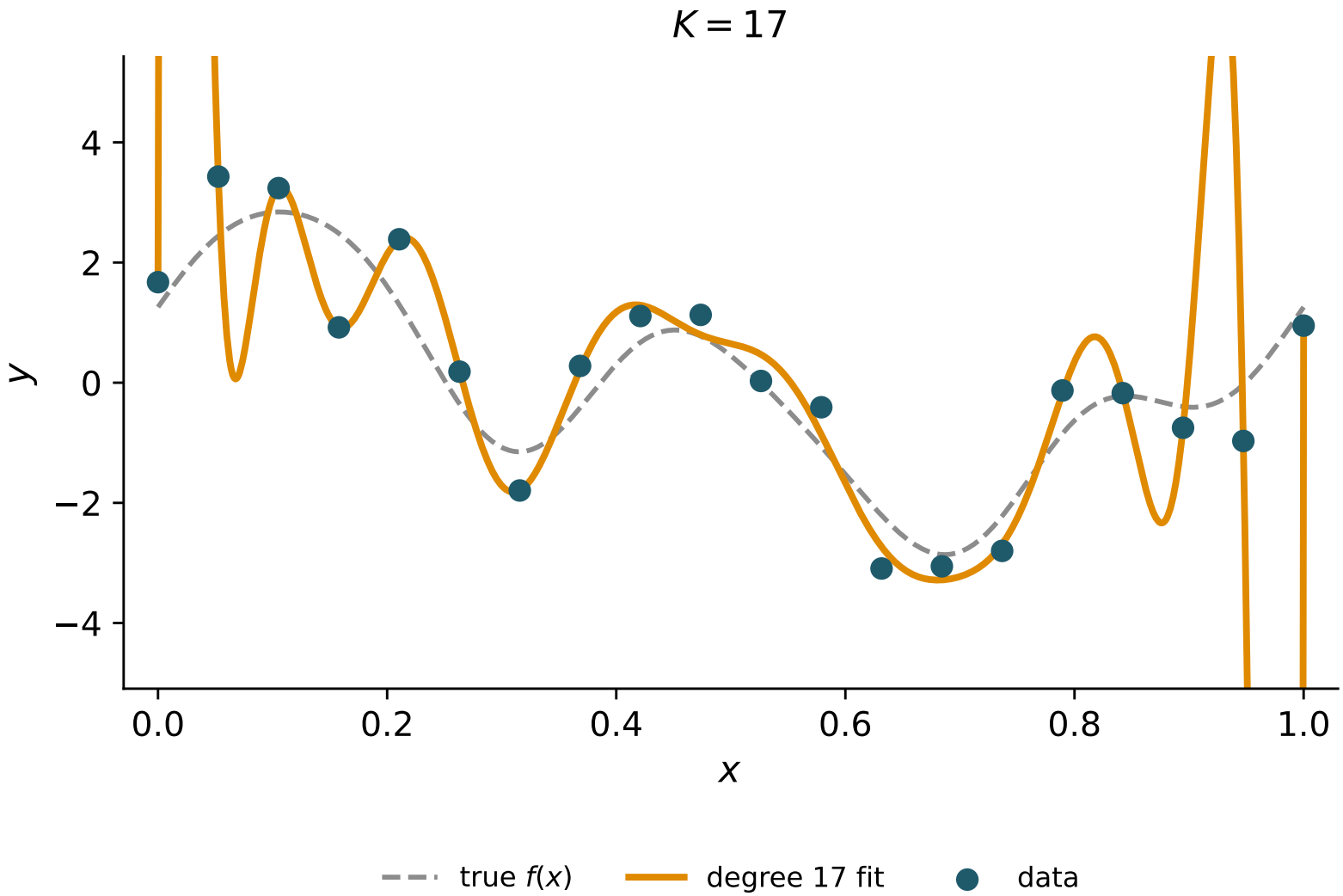
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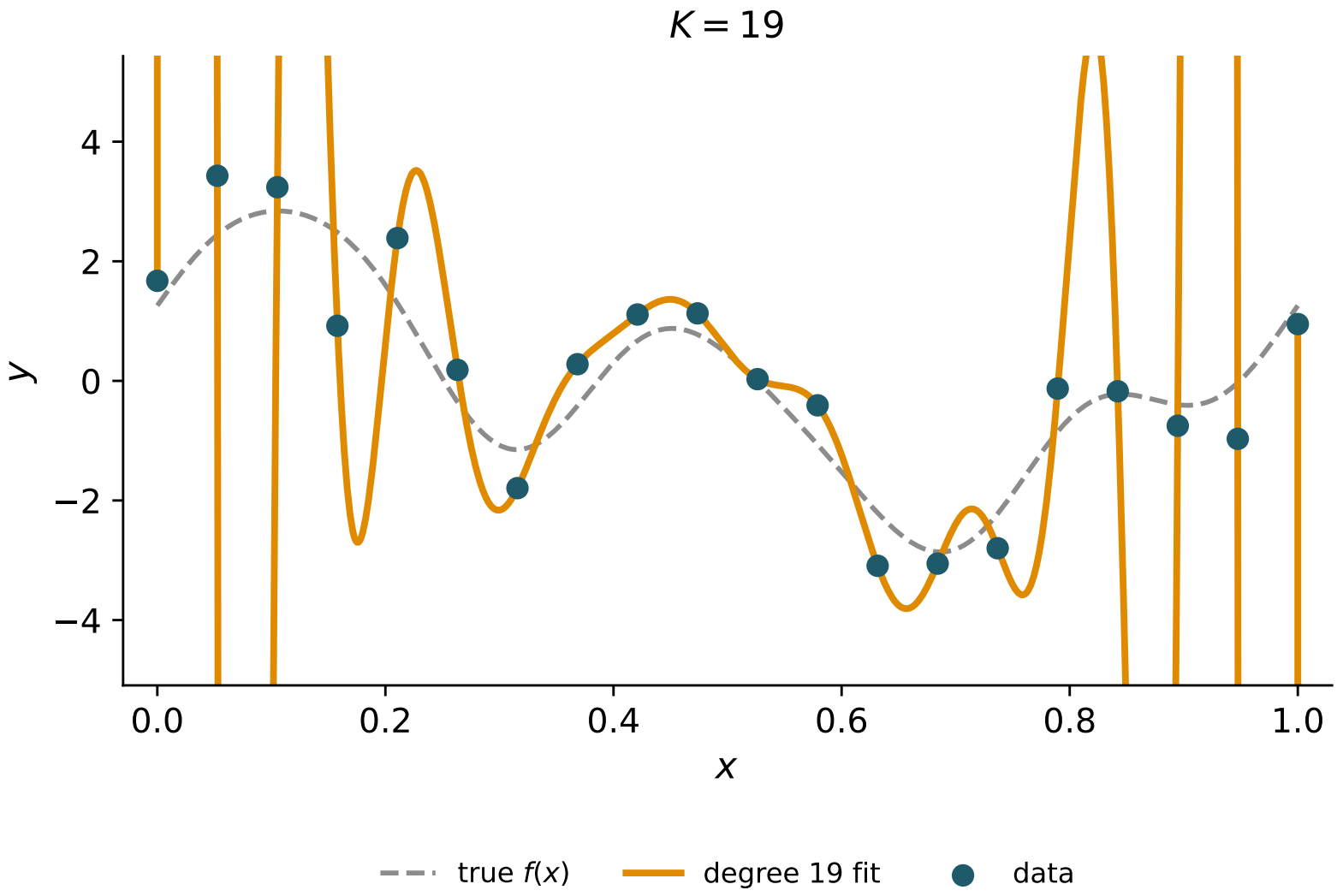
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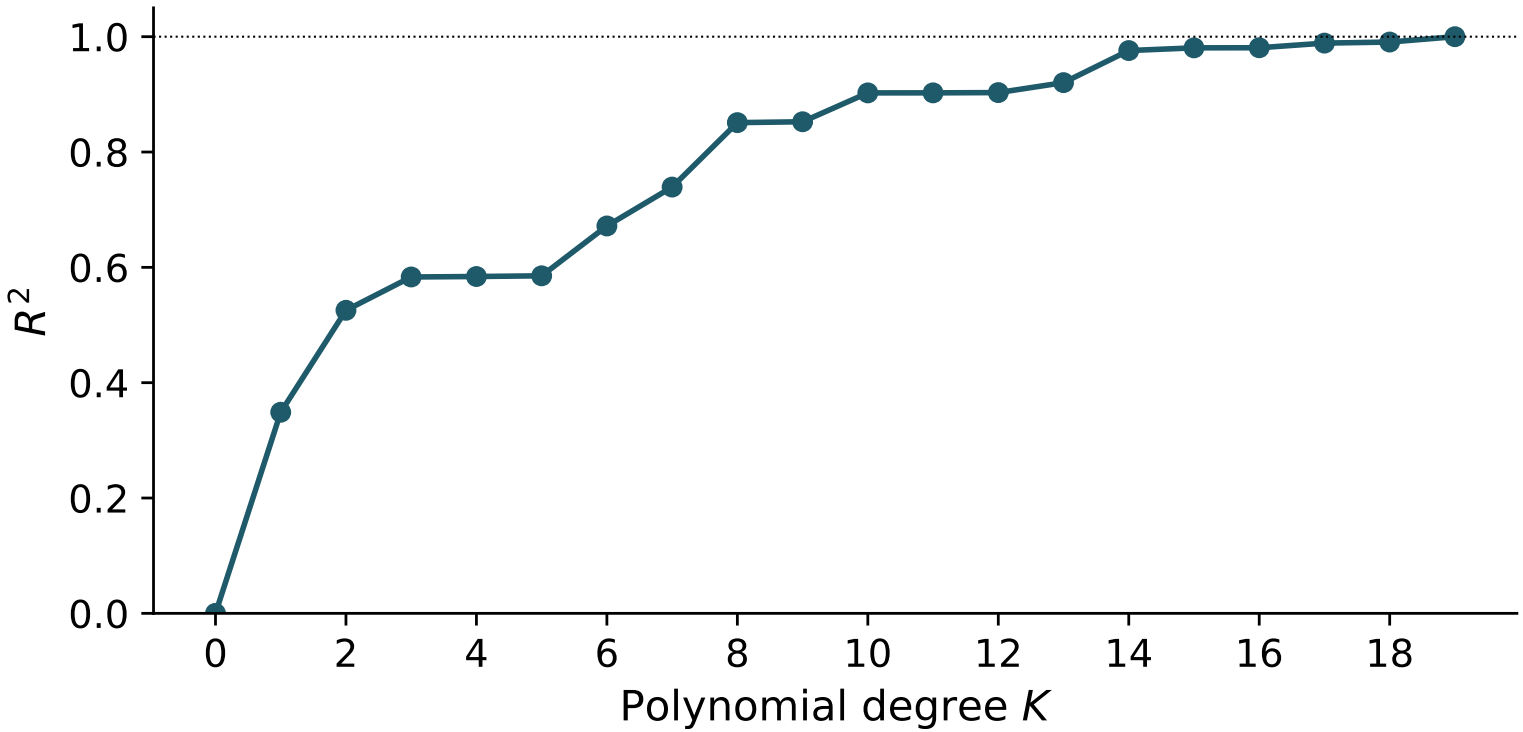


Example: polynomials fitted to a random function



R^2 as the degree increases

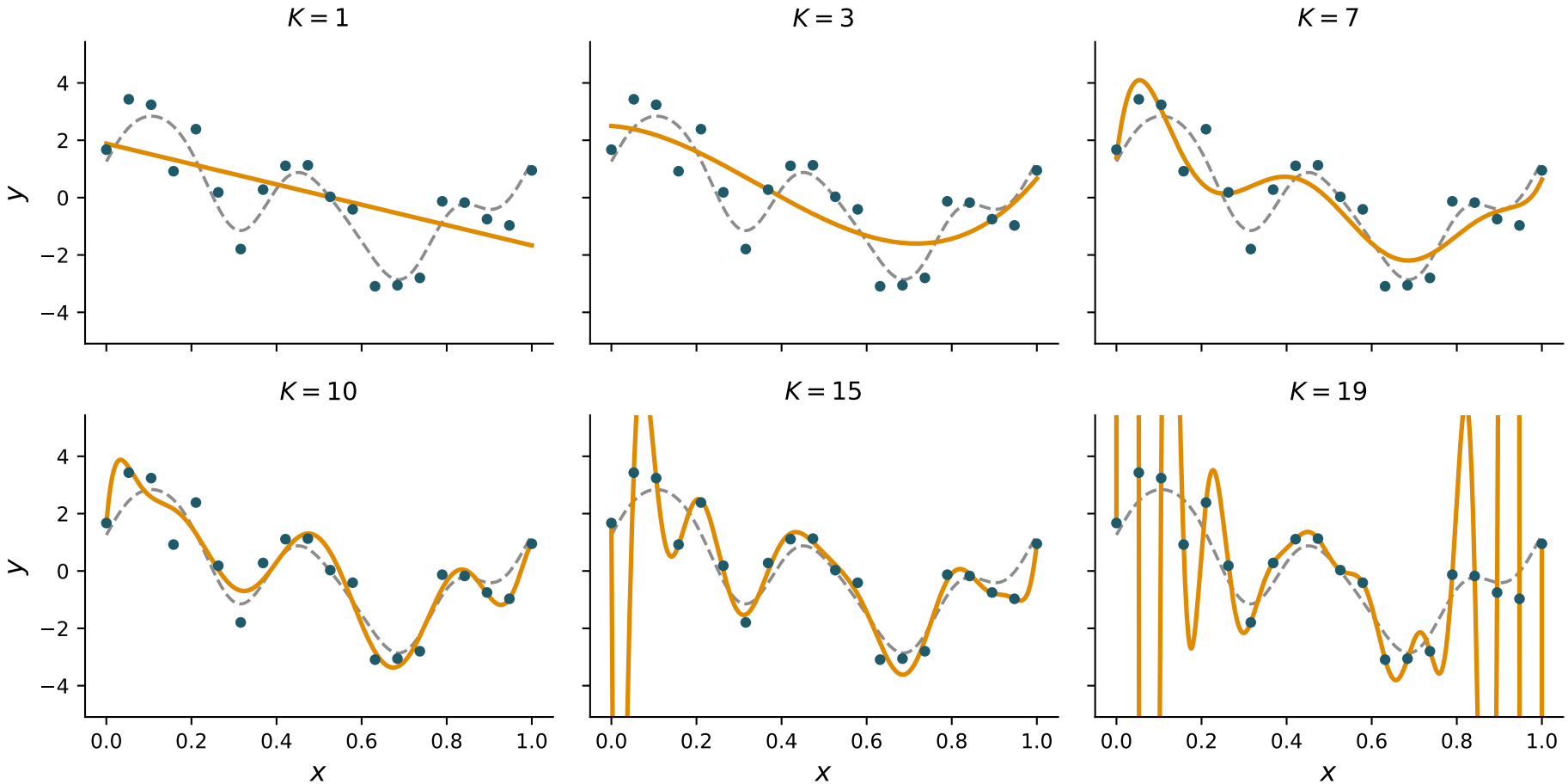
R^2 for the same 20 points and every degree $K = 0, 1, \ldots, 19$:



R^2 goes up with every coefficient we add and reaches 1 at $K = 19$, where the polynomial passes through all 20 points.

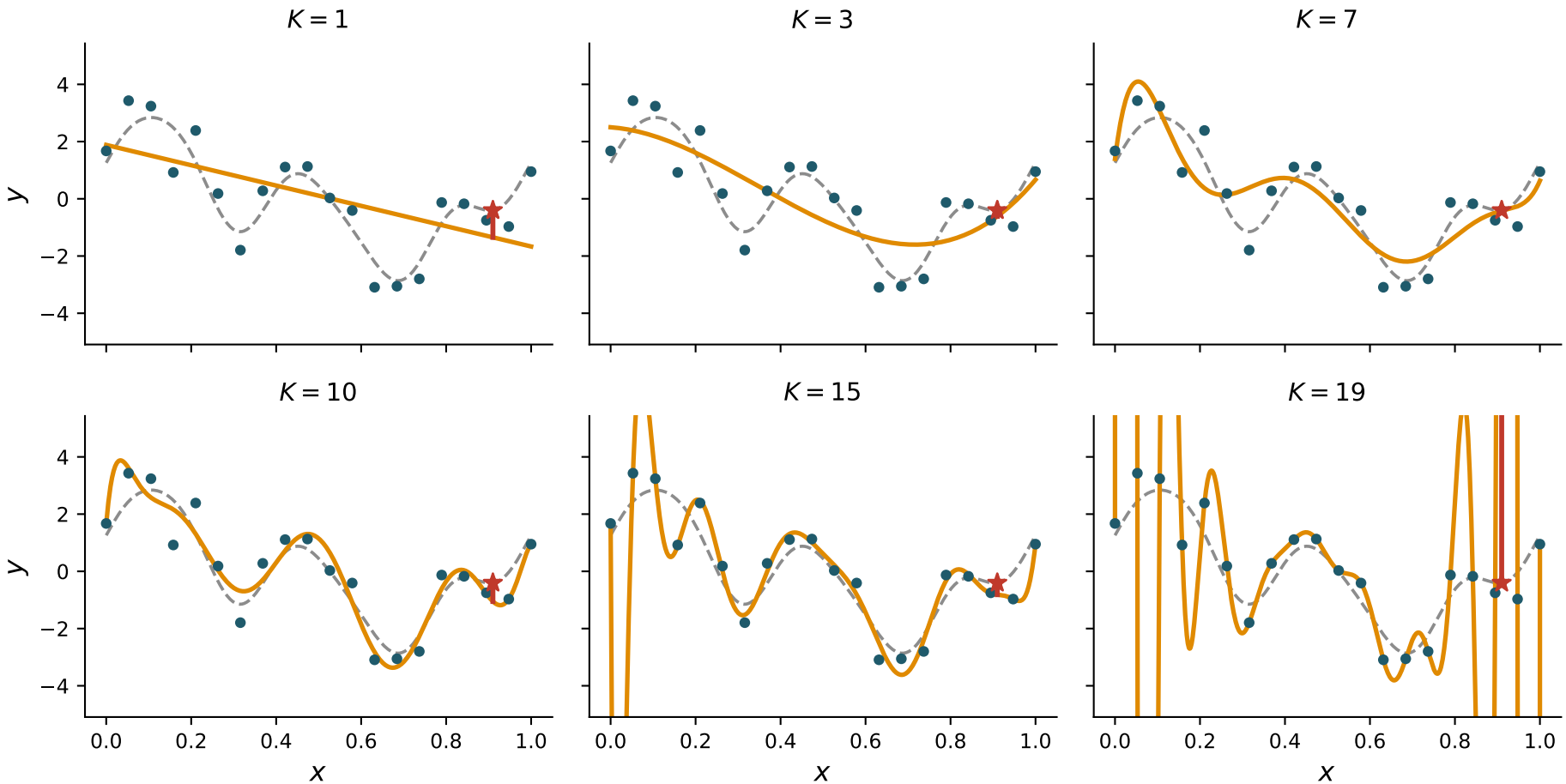
Predicting a new point

The true function (dashed), the 20 points and the fitted polynomials.



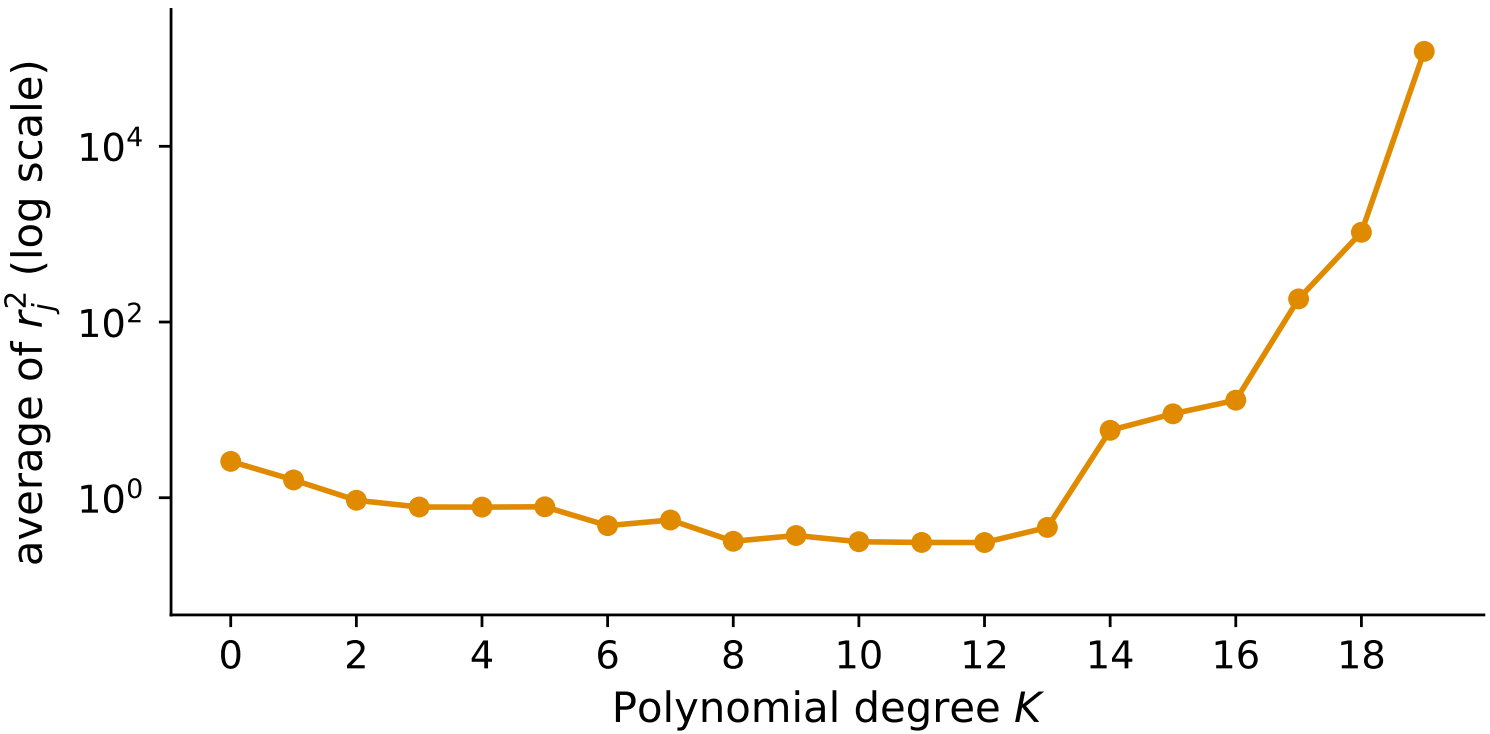
Predicting a new point

The true function (dashed), the 20 points and the fitted polynomials. A new point on the true function at $x = 0.91$ (star). Its residual $r = y - \hat{y}$ is the vertical distance from the fitted curve (red).



Average squared residual of the new points

Repeat this for 200 random points on the true function, with residuals r_j. For each degree K, the average of r_j^2:



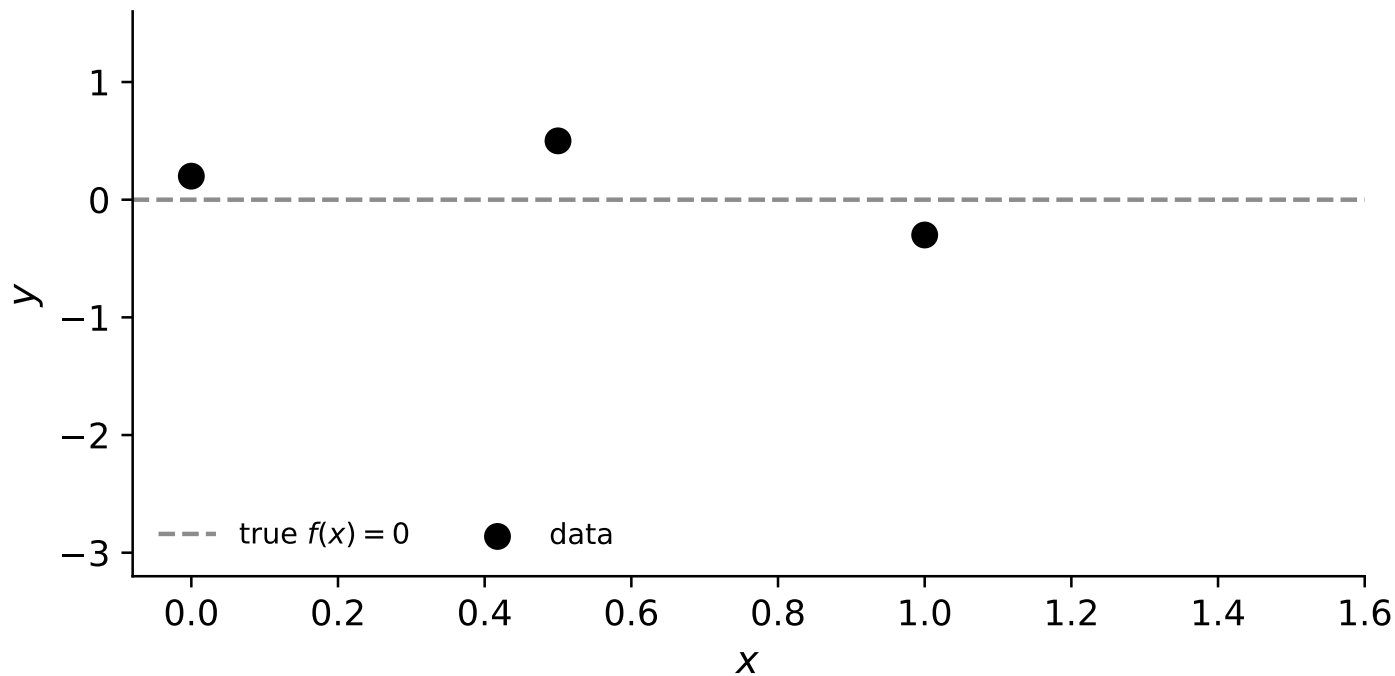
It is smallest around $K = 10$ to 12. For larger K it grows quickly, and at $K = 19$ it is about 120000.

Why the coefficients blow-up: minimal example

Consider the model

$$Y = \beta_0 + \beta_1 X + \beta_2 X^3 + \epsilon$$

Assume the data is generated with $\beta_0 = \beta_1 = \beta_2 = 0$, so there is no underlying relationship. Three data points, at $x = 0$, 0.5 and 1.

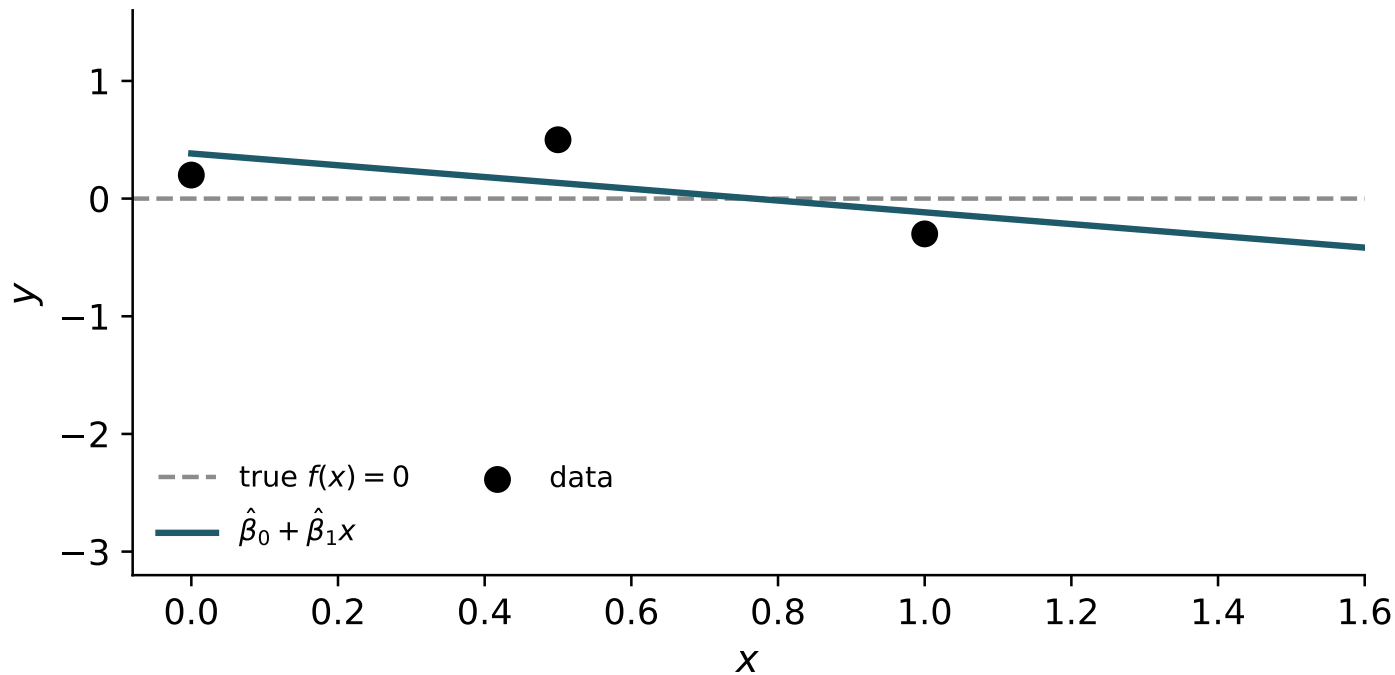


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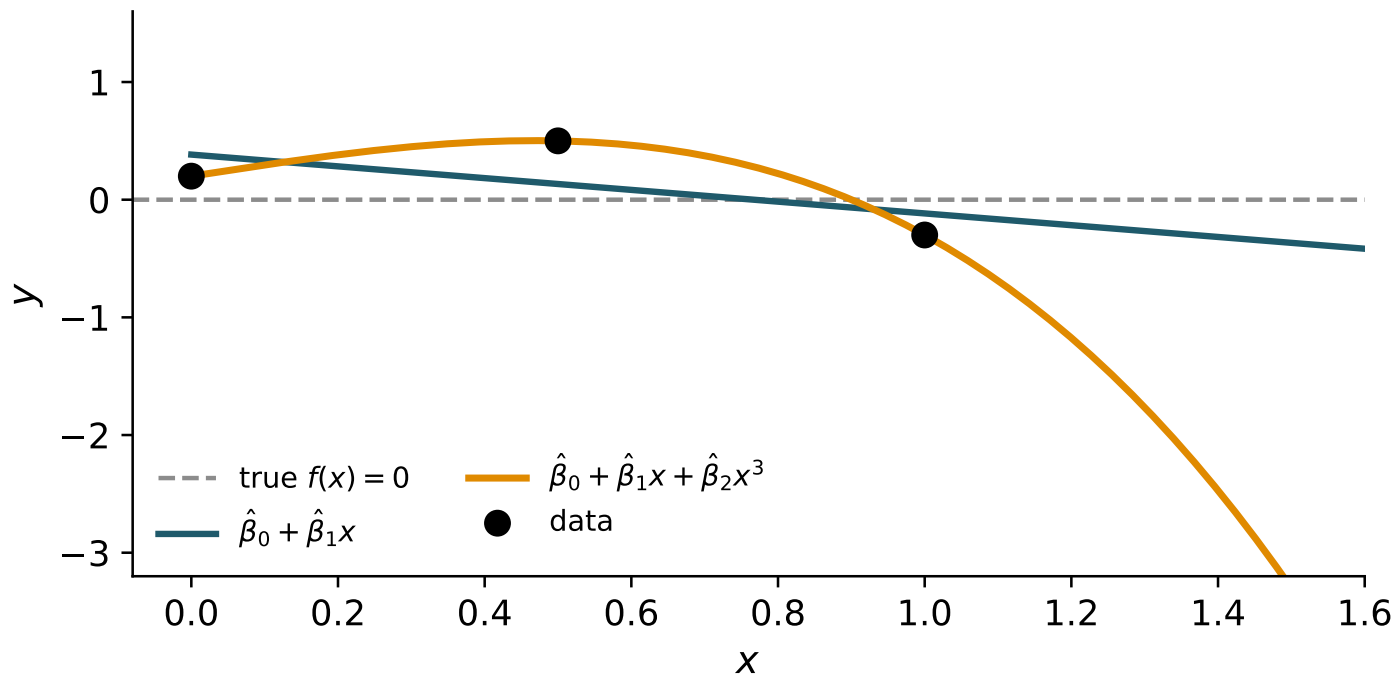
With only $\beta_0 + \beta_1 X$ (2 coefficients), the least-squares line cannot pass through all three points.

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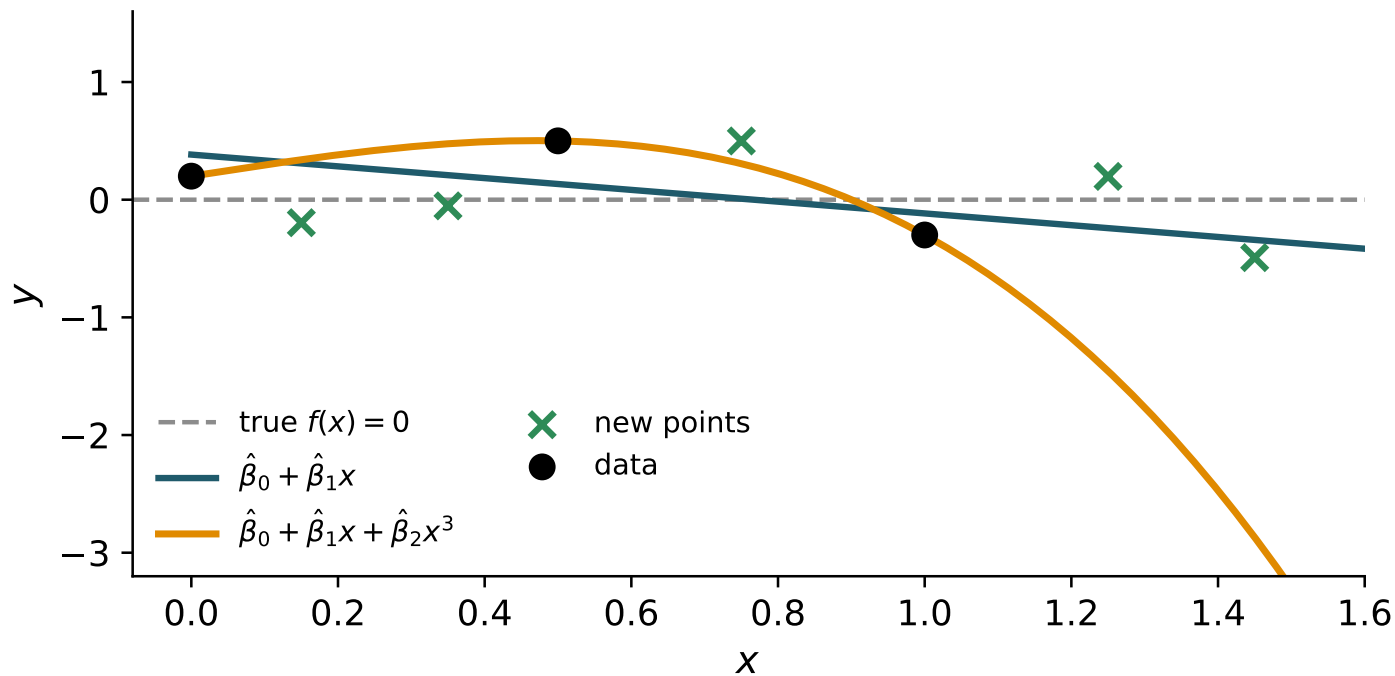


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Why are the coefficients so large?

The same model with all three coefficients 0 in reality, so $Y = \epsilon$. Three data points at $x = 0$, h and $2h$, with y values ϵ_0, ϵ_1 and ϵ_2. Three coefficients and three points, so the fitted curve passes through all of them:

$$\begin{aligned} \hat{\beta}_0 &= \epsilon_0 \\ \hat{\beta}_0 + \hat{\beta}_1 h + \hat{\beta}_2 h^3 &= \epsilon_1 \\ \hat{\beta}_0 + 2\hat{\beta}_1 h + 8\hat{\beta}_2 h^3 &= \epsilon_2 \end{aligned}$$

Question

Solve for $\hat{\beta}_2$. If ϵ_0, ϵ_1 and ϵ_2 are independent with variance σ_ϵ^2, what is $\mathrm{var}(\hat{\beta}_2)$? What happens to it as h gets smaller?

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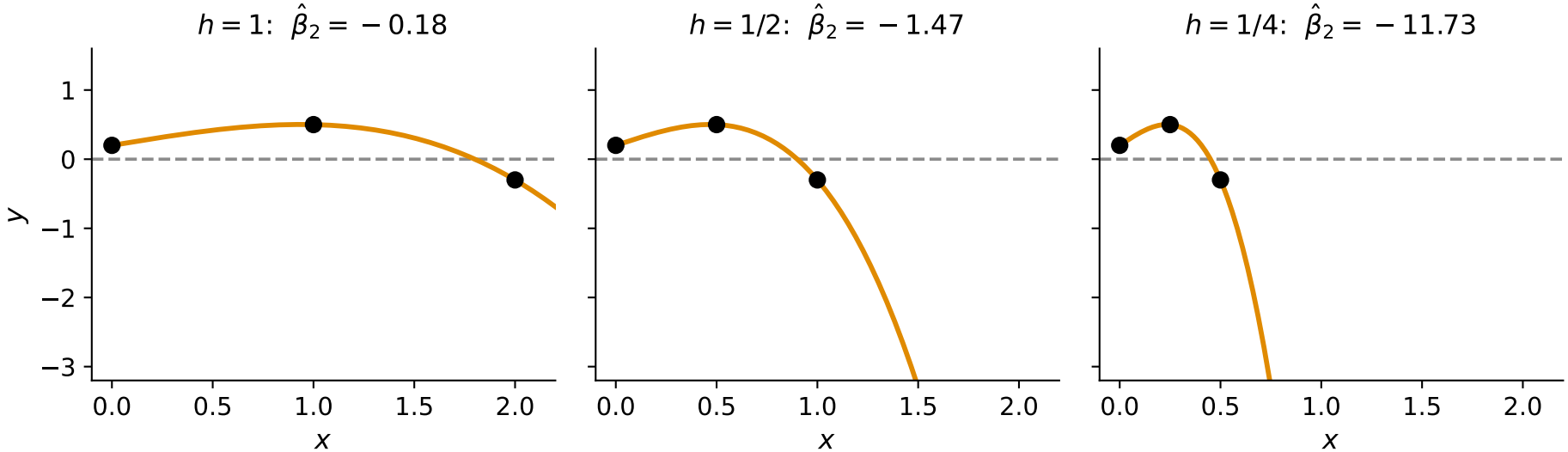
Put $\hat{\beta}_0 = \epsilon_0$ into the other two equations, then subtract twice the second from the third: $6\hat{\beta}_2 h^3 = \epsilon_2 - 2\epsilon_1 + \epsilon_0$, so

$$\hat{\beta}_2 = \frac{\epsilon_2 - 2\epsilon_1 + \epsilon_0}{6h^3}, \qquad \mathrm{var}(\hat{\beta}_2) = \frac{(1+4+1)\,\sigma_\epsilon^2}{36h^6} = \frac{\sigma_\epsilon^2}{6h^6}.$$

Halving h multiplies the standard deviation of $\hat{\beta}_2$ by $2^3 = 8$.

Why are the coefficients so large?

The same noise, $\epsilon_0 = 0.2$, $\epsilon_1 = 0.5$ and $\epsilon_2 = -0.3$, with the points closer and closer together:



Near 0, x^3 is tiny, so it takes a huge $\hat{\beta}_2$ to fit the noise. The degree-19 fit has the same problem with x^{19} and 20 points close together.